

Inequalities with one unknown

Solving an inequality is solving an equation with one extra rule — and an answer that is a whole interval, not a single number.

LESSON 49–50

2 × 40 MIN

ALGEBRA 8, §15

STAGE 9 · 4.3

LESSON CLOCK

10'

24'

16'

26'

4'

Warm-up	10 min
Explanation	24 min
Interactive	16 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- Solve a linear inequality in one unknown.
- Show the solution on a number line and write it as an interval.
- Reverse the sign when multiplying or dividing by a negative number.
- Recognise an inequality with no solutions or with every number as a solution.

TEXTBOOK

National	\$15, pp. 85–93
Cambridge	Learner’s Book pp. 96–102
Workbook	Workbook 4.3

01

Explanation

What a solution is

DEFINITION

A **solution** of an inequality with one unknown is any value of the unknown that makes the inequality a true statement. To **solve** the inequality is to find *all* of them.

An equation such as $2x = 6$ has one answer. The inequality $2x > 6$ has infinitely many — every number bigger than 3. So the answer is written as an **interval**, not a number.

The method

Exactly the steps you use for an equation, with one addition:

1. Clear fractions and open brackets.
2. Collect the unknowns on one side, the numbers on the other.

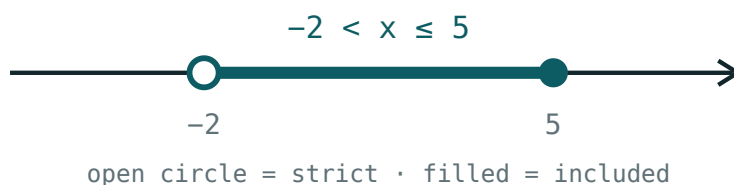
3. Divide by the coefficient of x — and if that coefficient is **negative**, **reverse the sign**.
4. Draw the answer on a number line and write the interval.

$$3x - 7 < x + 5 \implies 2x < 12 \implies x < 6$$

THE ONLY REAL TRAP

From $-4x \geq 12$ you get $x \leq -3$. Write the reversed sign **first**, then do the division — that way you cannot forget it.

Drawing and writing the answer



An open circle for a strict inequality, a filled circle when the endpoint is included.

INEQUALITY	ON THE LINE	INTERVAL
$x > 3$	open circle at 3, shaded right	$(3; +\infty)$
$x \geq 3$	filled circle at 3, shaded right	$[3; +\infty)$
$x < 3$	open circle at 3, shaded left	$(-\infty; 3)$
$x \leq 3$	filled circle at 3, shaded left	$(-\infty; 3]$

An infinity end is always written with a round bracket — you never reach it.

Two special answers

Sometimes the unknown disappears. Then read what is left:

$$2x + 5 > 2x + 1 \implies 5 > 1$$

That is true whatever x is, so **every number** is a solution: the answer is $(-\infty; +\infty)$.

$$3x + 2 < 3x - 4 \implies 2 < -4$$

That is false whatever x is, so the inequality has **no solutions**.

02 Worked examples

EXAMPLE 1 Solve $5x - 3 \leq 2x + 9$

1 $5x - 2x \leq 9 + 3$

Collect x on the left, numbers on the right.

2 $3x \leq 12$

3 $x \leq 4$

Dividing by 3, a positive number — the sign stays.

4 $(-\infty; 4]$

Filled circle at 4, shaded to the left.

ANSWER

$x \leq 4$, that is $(-\infty; 4]$

EXAMPLE 2 Solve $4 - 3x > 19$

- ① $-3x > 15$
Subtract 4 from both sides.
- ② Dividing by -3 — the sign must reverse.
Write the new sign first.
- ③ $x < -5$
- ④ $(-\infty; -5)$
Open circle at -5 .

ANSWER

$x < -5$

EXAMPLE 3 Solve $\frac{x-1}{2} + \frac{x}{3} \geq 1$

- ① multiply by 6
The LCD of 2 and 3 — 6 is positive, so the sign stays.
- ② $3(x-1) + 2x \geq 6$
- ③ $3x - 3 + 2x \geq 6$
- ④ $5x \geq 9$, so $x \geq 1.8$

ANSWER

$x \geq 1.8$, that is $[1.8; +\infty)$

03 Interactive model

Move the boundary, flip the sign, and toggle strict — three controls cover every answer in this lesson.

Solutions on the number line

$$x > 2$$

inequality $x > 2$ boundary **2** (open) $x = 0$ works? **no** $x = 6$ works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Inequality with one unknown	Bir noma'lumli tengsizlik	Неравенство с одним неизвестным
Solution of an inequality	Tengsizlikning yechimi	Решение неравенства
Solution set	Yechimlar to'plami	Множество решений
Interval	Oraliq	Промежуток
Number line	Sonlar o'qi	Числовая прямая
Equivalent inequalities	Teng kuchli tengsizliklar	Равносильные неравенства
Reverse the sign	Ishorani almashtirish	Изменить знак неравенства
No solutions	Yechimga ega emas	Нет решений
Every number	Har qanday son	Любое число

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$-2x > 10$ gives:

$x \geq 3$ as an interval is:

$2x + 5 > 2x + 1$ has:

$3x + 2 < 3x - 4$ has:

no solutions

one solution

every number

$x < 0$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Solve $2x > 8$*

ANSWER $x > 4$

2. *Solve $x + 3 \leq 7$*

ANSWER $x \leq 4$

3. *Solve $x - 5 > 1$*

ANSWER $x > 6$

4. *Solve $-x < 2$*

ANSWER $x > -2$

5. *Solve $3x \geq -9$*

ANSWER $x \geq -3$

6. *Solve $-2x \geq 6$*

ANSWER $x \leq -3$

7. *Write $x < 5$ as an interval*

ANSWER $(-\infty; 5)$

Medium · the standard question

7 PROBLEMS

1. Solve $5x - 3 \leq 2x + 9$

ANSWER $x \leq 4$

2. Solve $4 - 3x > 19$

ANSWER $x < -5$

3. Solve $2(x - 1) < x + 4$

ANSWER $x < 6$

4. Solve $\frac{x}{3} + 2 \geq 5$

ANSWER $x \geq 9$

5. Solve $7 - x \leq 2x + 1$

ANSWER $x \geq 2$

6. Write $x \geq -4$ as an interval

ANSWER $[-4; +\infty)$

7. Solve $-5x + 2 < 17$

ANSWER $x > -3$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{x-1}{2} + \frac{x}{3} \geq 1$

ANSWER $x \geq 1.8$

2. Solve $3(2x-1) - 2(x+4) < 5$

ANSWER $6x - 3 - 2x - 8 < 5$, so $x < 4$

3. Solve $2x + 5 > 2(x+1)$

ANSWER Every number — $5 > 2$ is always true.

4. Solve $3(x-2) < 3x-4$

ANSWER $3x - 6 < 3x - 4$ reduces to $-6 < -4$, which is always true — every number is a solution.

5. Find the smallest whole number satisfying $4x - 7 > 9$

ANSWER $x > 4$, so $x = 5$

6. Find the largest whole number satisfying $5 - 2x \geq -3$

ANSWER $x \leq 4$, so $x = 4$

7. For which x is $\frac{2x-1}{3} - \frac{x}{2} \leq 0$?

ANSWER Multiply by 6: $4x - 2 - 3x \leq 0$, so $x \leq 2$.

07 Homework

Homework — 6 problems

Algebra 8, §15, pp. 85–93. Draw the number line for every answer.

1. Solve $4x - 5 \geq x + 7$

2. Solve $6 - 5x > 21$

3. Solve $3(x+2) \leq 2x+10$

4. Solve $\frac{x}{4} - 1 < 2$

5. Find the smallest whole number with $7x - 3 > 25$

6. Solve $2(x - 3) > 2x - 5$ and explain your answer

Systems of inequalities.

Number intervals

Two conditions at once — the answer is where the two solution sets overlap.

LESSON 51–52

2 × 40 MIN

ALGEBRA 8, §16

STAGE 9 · 4.3

LESSON CLOCK

10'

24'

16'

26'

4'

Warm-up	10 min
Explanation	24 min
Interactive	16 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- Solve a system of two linear inequalities.
- Find the intersection of two intervals on the number line.
- Use the standard names and notation for intervals.
- Solve a double inequality such as $-2 < 3x + 1 \leq 10$.

TEXTBOOK

National	§16, pp. 94–104
Cambridge	Learner’s Book pp. 96–102
Workbook	Workbook 4.3

01

Explanation

The names of the intervals

CONDITION	NOTATION	NAME
$a \leq x \leq b$	$[a; b]$	segment (closed interval)
$a < x < b$	$(a; b)$	open interval
$a \leq x < b$	$[a; b)$	half-open interval
$x \geq a$	$[a; +\infty)$	closed ray
$x < a$	$(-\infty; a)$	open ray

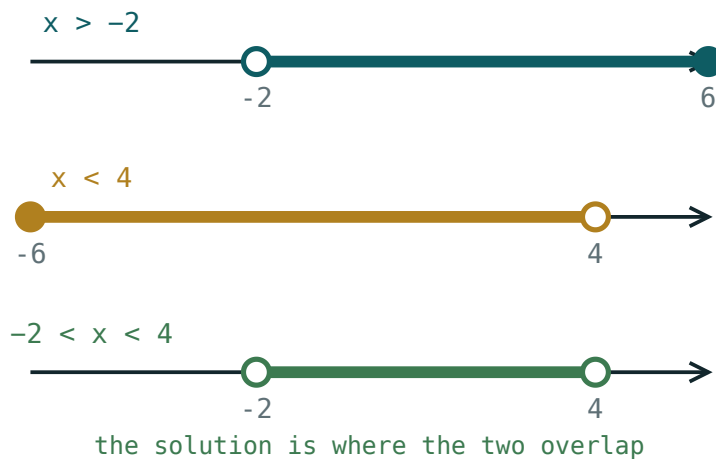
Square bracket = the end belongs to the set (filled circle). Round bracket = it does not (open circle). Infinity is always round.

Solving a system

METHOD

1. Solve each inequality on its own.

2. Draw both solution sets on the **same** number line.
3. The answer is the part shaded **twice** — the intersection.



Each inequality shades part of the line; the system is satisfied only where both shadings overlap.

The word to say in class is **“and”**. A system asks for the values satisfying the first inequality *and* the second, so only the overlap counts.

SOMETIMES THERE IS NO OVERLAP

The system $x > 5$ and $x < 2$ has **no solutions** — the empty set. Say so; do not write a nonsense interval like $(5; 2)$.

Double inequalities

$-2 < 3x + 1 \leq 10$ is a system written compactly. Work on all three parts at once:

$$-2 < 3x + 1 \leq 10$$

$$-3 < 3x \leq 9 \text{ (subtract 1 throughout)}$$

$$-1 < x \leq 3 \text{ (divide by 3 throughout)}$$

The answer is the half-open interval $(-1; 3]$.

IF YOU DIVIDE BY A NEGATIVE

Both signs reverse and the two ends swap places, so rewrite the smaller number on the left.

02 Worked examples

EXAMPLE 1 Solve the system: $2x - 1 > 3$ and $x + 4 < 9$

① $2x > 4$, so $x > 2$
The first inequality.

② $x < 5$
The second.

③ Draw both: shaded right of 2, and left of 5.

④ $2 < x < 5$
The overlap.

ANSWER

$(2; 5)$

EXAMPLE 2 Solve $-2 < 3x + 1 \leq 10$

① $-3 < 3x \leq 9$
Subtract 1 from all three parts.

② $-1 < x \leq 3$
Divide all three by 3.

③ $(-1; 3]$
Open at -1 , closed at 3.

ANSWER

$(-1; 3]$

EXAMPLE 3 Solve the system: $x \geq 4$ and $2x < 6$

① $x \geq 4$

The first.

② $x < 3$

The second.

③ Nothing is both at least 4 and less than 3.

④ No solutions — the empty set.

ANSWER

No solutions.

03 Interactive model

Set each boundary in turn and ask the class which part of the line survives both.

One inequality at a time

$x > 2$

inequality $x > 2$

boundary 2 (open)

 $x = 0$ works? no $x = 6$ works? yes

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
System of inequalities	Tengsizliklar sistemasi	Система неравенств
Intersection	Kesishma	Пересечение
Number interval	Sonli oraliq	Числовой промежуток
Segment (closed interval)	Kesma	Отрезок
Open interval	Ochiq oraliq	Интервал
Half-open interval	Yarim ochiq oraliq	Полуинтервал
Ray (half-line)	Nur	Луч
Double inequality	Qo‘sh tengsizlik	Двойное неравенство
Empty set	Bo‘sh to‘plam	Пустое множество

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$[2; 7)$ means:

The system $x > 1$ and $x < 6$ gives:

The system $x > 5$ and $x < 2$ gives:

$-4 \leq 2x < 6$ gives:

$[-2; 3)$

$(-2; 3]$

$[-4; 6)$

$[-2; 3]$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Write $3 \leq x \leq 8$ as an interval

ANSWER $[3; 8]$

2. Write $0 < x < 5$ as an interval

ANSWER $(0; 5)$

3. Write $x \geq 2$ as an interval

ANSWER $[2; +\infty)$

4. What does $(-1; 4]$ mean?

ANSWER $-1 < x \leq 4$

5. Solve the system $x > 1$ and $x < 6$

ANSWER $(1; 6)$

6. Solve the system $x \geq 0$ and $x \leq 3$

ANSWER $[0; 3]$

7. Solve the system $x > 4$ and $x > 7$

ANSWER $(7; +\infty)$

Medium · the standard question

7 PROBLEMS

1. Solve: $2x - 1 > 3$ and $x + 4 < 9$

ANSWER $(2; 5)$

2. Solve: $3x \leq 12$ and $x + 1 > 0$

ANSWER $(-1; 4]$

3. Solve $-2 < 3x + 1 \leq 10$

ANSWER $(-1; 3]$

4. Solve $-4 \leq 2x < 6$

ANSWER $[-2; 3)$

5. Solve: $x \geq 4$ and $2x < 6$

ANSWER No solutions.

6. Solve $1 \leq x - 3 \leq 5$

ANSWER $[4; 8]$

7. Solve: $5 - x > 1$ and $x + 2 \geq 0$

ANSWER $[-2; 4)$

Hard · stretch and reason

7 PROBLEMS

1. Solve $-5 < 2 - 3x \leq 8$

ANSWER $-7 < -3x \leq 6 \implies -2 \leq x < \frac{7}{3}$

2. Solve: $2(x - 1) \geq x$ and $3x < x + 8$

ANSWER $x \geq 2$ and $x < 4$: $[2; 4)$

3. How many whole numbers satisfy $-3 < 2x + 1 \leq 9$?

ANSWER $-2 < x \leq 4$: the numbers $-1, 0, 1, 2, 3, 4$ — six of them.

4. Solve: $\frac{x}{2} > 1$ and $\frac{x}{3} < 2$

ANSWER $x > 2$ and $x < 6$: $(2; 6)$

5. For which a does the system $x > 3$ and $x < a$ have solutions?

ANSWER $a > 3$

6. Solve $0 \leq \frac{2x - 1}{3} \leq 3$

ANSWER $0 \leq 2x - 1 \leq 9 \implies [0.5; 5]$

7. Find all whole numbers satisfying both $4x > 6$ and $3x \leq 15$

ANSWER $x > 1.5$ and $x \leq 5$: 2, 3, 4, 5

07 Homework

Homework — 6 problems

Algebra 8, §16, pp. 94–104. Draw both sets on one line before writing the answer.

- Solve: $3x - 2 > 7$ and $x - 1 < 6$
- Solve $-6 \leq 4x + 2 < 10$
- Solve: $x + 5 \geq 2$ and $2x \leq 8$
- Solve: $x > 6$ and $x < 3$, and explain the answer

5. *How many whole numbers satisfy $-4 < 3x - 1 \leq 11$?*
6. *Write $[-3; 5)$ in words and draw it on a number line*

The modulus of a number

Distance from zero — one idea that turns every modulus equation and inequality into something you can already solve.

LESSON 53–54

2 × 40 MIN

ALGEBRA 8, §17

STAGE 9 · BEYOND

LESSON CLOCK

10'

24'

16'

26'

4'

Warm-up	10 min
Explanation	24 min
Interactive	16 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- Define $|a|$ and evaluate it.
- Read $|x|$ as the distance from x to 0 on the number line.
- Solve equations of the form $|x - a| = b$.
- Solve inequalities $|x| < b$ and $|x| > b$.

TEXTBOOK

National	\$17, pp. 105–110
Cambridge	Extension beyond Stage 9
Workbook	—

01

Explanation

The definition

DEFINITION

$|a| = a$ if $a \geq 0$, $|a| = -a$ if $a < 0$

So $|5| = 5$, $|-5| = 5$, $|0| = 0$. The modulus is **never negative**.

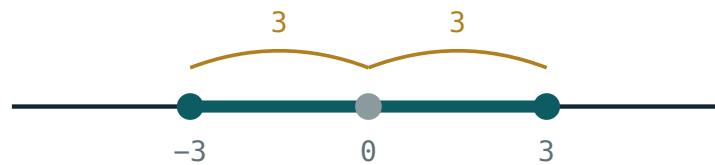
-A IS NOT A NEGATIVE NUMBER

If $a = -5$ then $-a = 5$. The second line of the definition *removes* the minus sign; it does not add one.

Two facts worth stating at once: $|a| \geq 0$ always, and $|a| = |-a|$ — opposite numbers have the same modulus.

The picture: distance from zero

$|a|$ is the **distance** from the point a to the origin. That single sentence solves the whole topic.



$$|x| \leq 3 \text{ means } -3 \leq x \leq 3$$

Every point within 3 of the origin satisfies $|x| \leq 3$ —
the segment from -3 to 3 .

More generally, $|x - a|$ is the distance from x to a . So $|x - 4| = 3$ asks: which points are exactly 3 away from 4? Answer: 1 and 7.

Equations

RULE

For $b > 0$: $|A| = b$ means $A = b$ **or** $A = -b$.

$|A| = 0$ means $A = 0$. $|A| = b$ with $b < 0$ has **no solutions**.

$$|x - 4| = 3 \implies x - 4 = 3 \text{ or } x - 4 = -3 \implies x = 7 \text{ or } x = 1$$

Always check the sign of the right-hand side first. $|x + 2| = -5$ needs no working at all — a modulus cannot be negative.

Inequalities

TWO SHAPES, TWO ANSWERS

$|x| < b$ means $-b < x < b$ — one interval, in the middle

$|x| > b$ means $x < -b$ **or** $x > b$ — two rays, outside

The picture makes it obvious: “less than b from zero” is the middle piece; “more than b from zero” is everything outside.

Same with a shift: $|x - 5| \leq 2$ means the distance from 5 is at most 2, so $3 \leq x \leq 7$.

02 Worked examples

EXAMPLE 1 Solve $|2x - 1| = 7$

① $2x - 1 = 7$ or $2x - 1 = -7$

Two cases.

② $2x = 8$, so $x = 4$

First case.

③ $2x = -6$, so $x = -3$

Second case.

④ check: $|7| = 7$ ✓ and $|-7| = 7$ ✓

ANSWER

$x = 4$ or $x = -3$

EXAMPLE 2 Solve $|x - 3| < 5$

① The distance from 3 is less than 5.

Read it as distance.

② $-5 < x - 3 < 5$

The middle-piece form.

③ $-2 < x < 8$

Add 3 throughout.

ANSWER

$(-2; 8)$

EXAMPLE 3 Solve $|x + 1| \geq 4$

- ① The distance from -1 is at least 4.
This is the “outside” shape.

- ② $x + 1 \leq -4$ or $x + 1 \geq 4$
Two rays.

- ③ $x \leq -5$ or $x \geq 3$

ANSWER

$$(-\infty; -5] \cup [3; +\infty)$$

03 Interactive model

Set the boundary at 3 and switch between $<$ and $>$ — the two answer shapes appear side by side.

Inside or outside?

$$x > 2$$

inequality $x > 2$

boundary 2 (open)

 $x = 0$ works? no $x = 6$ works? yes**04** Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Modulus (absolute value)	Modul (absolyut qiymat)	Модуль (абсолютная величина)
Distance from zero	Noldan uzoqlik	Расстояние от нуля
Non-negative	Manfiy bo'lmagan	Неотрицательный
Two cases	Ikki hol	Два случая
Opposite numbers	Qarama-qarshi sonlar	Противоположные числа
Equation with a modulus	Modulli tenglama	Уравнение с модулем
Inequality with a modulus	Modulli tengsizlik	Неравенство с модулем
Solution set	Yechimlar to'plami	Множество решений

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$|-7|$ equals:

-7

7

0

± 7

$|x| = 5$ gives:

$x = 5$

$x = -5$

$x = 5$ or $x = -5$

no solutions

$|x| < 4$ means:

$x < 4$

$-4 < x < 4$

$x > 4$

$x < -4$ or $x > 4$

$|x + 2| = -3$ has:

one solution

two solutions

no solutions

every number

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find $|9|$

ANSWER 9

2. Find $|-9|$

ANSWER 9

3. Find $|0|$

ANSWER 0

4. Find $|-3| + |4|$

ANSWER 7

5. Solve $|x| = 6$

ANSWER $x = 6$ or $x = -6$

6. Solve $|x| = 0$

ANSWER $x = 0$

7. Solve $|x| < 2$

ANSWER $-2 < x < 2$

Medium · the standard question

7 PROBLEMS

1. Solve $|x - 4| = 3$

ANSWER $x = 7$ or $x = 1$

2. Solve $|2x - 1| = 7$

ANSWER $x = 4$ or $x = -3$

3. Solve $|x + 2| = 5$

ANSWER $x = 3$ or $x = -7$

4. Solve $|x - 3| < 5$

ANSWER $(-2; 8)$

5. Solve $|x| \geq 3$

ANSWER $x \leq -3$ or $x \geq 3$

6. Solve $|x + 1| \geq 4$

ANSWER $x \leq -5$ or $x \geq 3$

7. Solve $|x + 2| = -5$

ANSWER No solutions.

Hard · stretch and reason

7 PROBLEMS

1. Solve $|3x + 6| = 12$

ANSWER $x = 2$ or $x = -6$

2. Solve $|x - 5| \leq 2$

ANSWER $[3; 7]$

3. Solve $|2x - 3| < 5$

ANSWER $-1 < x < 4$

4. Solve $|1 - x| = 4$

ANSWER $x = -3$ or $x = 5$

5. How many whole numbers satisfy $|x| \leq 4$?

ANSWER Nine: -4 to 4 .

6. Solve $|x - 2| > 0$

ANSWER Every x except $x = 2$.

7. Find the distance between the points $x = -3$ and $x = 8$ using a modulus

ANSWER $|8 - (-3)| = 11$

07 Homework

Homework — 6 problems

Algebra 8, §17, pp. 105–110. Draw the number line for the inequalities.

1. Find $|-12|$, $|12|$ and $|-12| - |5|$

2. Solve $|x - 6| = 4$

3. Solve $|3x + 1| = 10$

4. Solve $|x| \leq 7$

5. Solve $|x - 1| > 3$

6. *Explain why $|x + 3| = -2$ has no solutions*

Approximate values and error

Every measurement is approximate — this lesson says by how much, and how to write it honestly.

LESSON 55–57

3 × 40 MIN

ALGEBRA 8, §§18–19

STAGE 9 · 3.4

LESSON CLOCK

15'

36'

24'

39'

6'

Warm-up	15 min
Explanation	36 min
Interactive	24 min
Practice	39 min
Homework	6 min

LEARNING OBJECTIVES

- Distinguish an exact value from an approximate one.
- Calculate the absolute error and the relative error.
- Round to a given number of decimal places or significant figures.
- Say which of two measurements is the more accurate.

TEXTBOOK

National	\$18 pp. 111–113, \$19 pp. 114–117
Cambridge	Learner’s Book pp. 74–81
Workbook	Workbook 3.4

01

Explanation

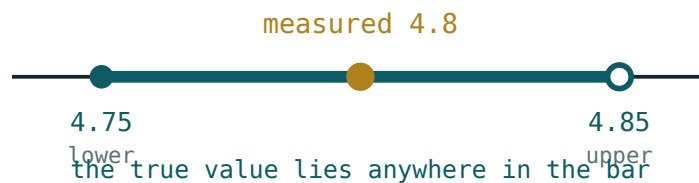
Exact and approximate

A count is exact: there are exactly 27 learners in the class. A measurement is not: a table is “1.2 m long” only to the accuracy of the tape. Every measured number is an **approximate value** of some exact value we shall never know.

DEFINITION

If a is the exact value and x an approximation of it, the **absolute error** is $\Delta = |a - x|$.

We rarely know Δ exactly — we know a **bound** for it. Writing $x = 4.8 \pm 0.05$ says the true value lies between 4.75 and 4.85.



*A measurement of 4.8 rounded to one decimal place:
the true value lies anywhere in the bar.*

Relative error

An error of 1 cm matters enormously on a pencil and not at all on a road. What decides is the error **compared with the size** of the thing measured.

DEFINITION

$$\text{relative error} = \frac{\Delta}{|x|}, \text{ usually written as a percentage}$$

1 cm on a 20 cm pencil is $\frac{1}{20} = 5\%$; 1 cm on a 1000 cm road is 0.1% . The second measurement is a hundred times more accurate, even though the absolute error is the same.

TO COMPARE TWO MEASUREMENTS

Compare the **relative** errors. The smaller relative error is the more accurate measurement.

Rounding

To a decimal place. Look at the first digit you are dropping: 5 or more, round up; less than 5, leave it. $3.147 \rightarrow 3.15$ to 2 d.p.

To significant figures. Count from the first non-zero digit. $0.004736 \rightarrow 0.0047$ to 2 s.f.; $52\,800 \rightarrow 53\,000$ to 2 s.f.

THE ERROR A ROUNDING CREATES

Rounding to the nearest 0.1 gives an absolute error of at most 0.05 — half the last unit. To the nearest whole number, at most 0.5 .

ROUND ONCE, AT THE END

Rounding part-way through a calculation and then again at the end makes the error grow. Keep the full figures until the last line.

02 Worked examples

EXAMPLE 1 The exact value is $a = 3.14159$ and the approximation is $x = 3.14$. Find both errors.

① $\Delta = |3.14159 - 3.14| = 0.00159$

Absolute error.

② $\frac{0.00159}{3.14} \approx 0.00051$

Divide by the approximation.

③ $\approx 0.05\%$

As a percentage.

ANSWER

$\Delta \approx 0.0016$, relative error $\approx 0.05\%$

EXAMPLE

2

A length is measured as 4.8 m to one decimal place. Between which values does the true length lie?

① One decimal place means the nearest 0.1 .

② The error is at most half of that: 0.05 .

③ $4.8 - 0.05 = 4.75$ and $4.8 + 0.05 = 4.85$

④ $4.75 \leq \text{length} < 4.85$

The lower bound is included, the upper is not.

ANSWER

$$4.75 \text{ m} \leq \ell < 4.85 \text{ m}$$

EXAMPLE

3

Which is more accurate: 12 cm measured to the nearest cm, or 250 cm measured to the nearest cm?

① Both have the same absolute error, at most 0.5 cm .

② $\frac{0.5}{12} \approx 4.2\%$

First relative error.

③ $\frac{0.5}{250} = 0.2\%$

Second relative error.

④ The second is far more accurate.

The same error on a bigger quantity matters less.

ANSWER

The 250 cm measurement.

03 Interactive model

Ask for a bound before any calculation — “the answer is somewhere between ...”.

Errors and rounding

Exact $a = 3.14159$, approximation $x = 3.14$.

$$① \quad \Delta = |a - x| = 0.00159$$

Absolute
error

—
just
the
difference,
without
a
sign.

$$② \quad \frac{\Delta}{x} = \frac{0.00159}{3.14} \approx 0.00051$$

Relative
error.

$$③ \quad \approx 0.05\%$$

Multiply
by
100
to
state
it as
a
percentage.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Exact value	Aniq qiymat	Точное значение
Approximate value	Taqribiy qiymat	Приближённое значение
Absolute error	Absolyut xatolik	Абсолютная погрешность
Relative error	Nisbiy xatolik	Относительная погрешность
Round	Yaxlitlash	Округление
Decimal place	O'nli xona	Десятичный знак
Significant figures	Ahamiyatli raqamlar	Значащие цифры
Accuracy	Aniqlik	Точность
Measurement	O'lchash	Измерение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The absolute error of an approximation is:

$$a - x$$

$$|a - x|$$

$$\frac{a}{x}$$

$$a + x$$

A length rounded to 4.8 m (1 d.p.) lies between:

4.7 and 4.9

4.75 and 4.85

4.79 and 4.81

4.8 exactly

0.004736 to 2 s.f. is:

0.00

0.0047

0.0048

0.004

To compare the accuracy of two measurements you use:

- the absolute error
- the relative error
- the larger value
- the number of digits

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Round 3.847 to 1 d.p.

ANSWER 3.8

2. Round 3.847 to 2 d.p.

ANSWER 3.85

3. Round 5629 to 2 s.f.

ANSWER 5600

4. Round 0.00382 to 2 s.f.

ANSWER 0.0038

5. Exact 7, approximation 6.9. Find the absolute error.

ANSWER 0.1

6. Exact 50, approximation 49. Find the absolute error.

ANSWER 1

7. A length is 6 m to the nearest metre. Find the largest possible error.

ANSWER 0.5 m

Medium · the standard question

7 PROBLEMS

1. *Exact 3.14159, approximation 3.14. Find the absolute error.*

ANSWER ≈ 0.0016

2. *The same numbers: find the relative error as a percentage.*

ANSWER $\approx 0.05\%$

3. *A length reads 4.8 m to 1 d.p. Give the bounds.*

ANSWER $4.75 \leq \ell < 4.85$

4. *A mass reads 250 g to the nearest 10 g. Give the bounds.*

ANSWER $245 \leq m < 255$

5. *Exact 20, approximation 19.5. Find the relative error.*

ANSWER 2.5%

6. *Which is more accurate: 12 cm or 250 cm, both to the nearest cm?*

ANSWER 250 cm — relative error 0.2% against 4.2%.

7. *Round 0.0004736 to 3 s.f.*

ANSWER 0.000474

Hard · stretch and reason

7 PROBLEMS

1. A rectangle is 4.2 m by 3.5 m, both to 1 d.p. Find the bounds of its perimeter.

ANSWER $4.15 \leq a < 4.25$, $3.45 \leq b < 3.55$, so $15.2 \leq P < 15.6$

2. The same rectangle: find the bounds of its area.

ANSWER $14.3175 \leq S < 15.0875$

3. A measurement has absolute error at most 0.02 and relative error 0.5%. Estimate the value.

ANSWER $\frac{0.02}{x} = 0.005$, so $x = 4$

4. Two masses: $3.5 \text{ kg} \pm 0.05$ and $480 \text{ g} \pm 5 \text{ g}$. Which is more accurate?

ANSWER 1.43% against 1.04% — the 480 g measurement.

5. Explain why rounding twice can be worse than rounding once.

ANSWER Each rounding adds its own error, and the errors accumulate: $2.46 \rightarrow 2.5 \rightarrow 3$ but $2.46 \rightarrow 2$ directly.

6. A number rounds to 7.30 to 2 d.p. Give its bounds.

ANSWER $7.295 \leq x < 7.305$

7. A speed of 60 km/h is measured to the nearest 5 km/h. Find the relative error at most.

ANSWER $\frac{2.5}{60} \approx 4.2\%$

07 Homework

Homework — 6 problems

Algebra 8, §§18–19, pp. 111–117.

1. Round 2.7182 to 1, 2 and 3 decimal places.
2. Round 34 782 to 2 and to 3 significant figures.
3. Exact 8, approximation 7.6. Find the absolute and relative errors.
4. A length reads 12.3 cm to 1 d.p. Give the bounds.

5. Which is more accurate: 5 m to the nearest metre, or 500 m to the nearest metre?
6. A square has side 6.4 cm to 1 d.p. Find the bounds of its perimeter.

Upper and lower bounds

The national revision and practical lessons, carrying the Cambridge Stage 9 treatment of bounds in calculations.

LESSON 58–59

2 × 40 MIN

ALGEBRA 8, CHAPTER II REVISION

STAGE 9 · 3.4 [CAMBRIDGE INSERT]

LESSON CLOCK

10'

24'

16'

26'

4'

Warm-up	10 min
Explanation	24 min
Interactive	16 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- Write the upper and lower bound of a rounded measurement.
- Find the bounds of a sum and of a difference.
- Find the bounds of a product and of a quotient.
- Decide how many figures an answer may honestly be given to.

TEXTBOOK

National	pp. 118–121
Cambridge	Learner’s Book pp. 76–81
Workbook	Workbook 3.4

01

Explanation

The bounds of a rounded number

RULE

A value rounded to the nearest u lies within $\frac{u}{2}$ of the stated figure:

$$x - \frac{u}{2} \leq \textit{true value} < x + \frac{u}{2}$$

STATED	ROUNDED TO	LOWER BOUND	UPPER BOUND
7 cm	nearest cm	6.5	7.5
4.8 m	1 d.p.	4.75	4.85
250 g	nearest 10 g	245	255
3.60 s	2 d.p.	3.595	3.605

The lower bound is reached; the upper bound is not. That is why it is written with $\leq$ on the left and $<$ on the right.

Bounds of a sum, difference, product, quotient

WHICH BOUND WITH WHICH		
	LARGEST ANSWER	SMALLEST ANSWER
$a + b$	$UB(a) + UB(b)$	$LB(a) + LB(b)$
$a - b$	$UB(a) - LB(b)$	$LB(a) - UB(b)$
$a \times b$	$UB(a) \times UB(b)$	$LB(a) \times LB(b)$
$a \div b$	$UB(a) \div LB(b)$	$LB(a) \div UB(b)$

(for positive quantities)

The pattern is worth saying aloud: for a **sum** or a **product**, both go the same way; for a **difference** or a **quotient**, they go opposite ways — to make a difference as large as possible, take the most you can and subtract the least you can.

How many figures may the answer have?

Work out both bounds, then keep only the digits on which they agree.

If the bounds of an area come out as 14.3175 and 15.0875, the two do not even agree on the units digit — so the honest answer is “about 15”, not 14.7 m^2 . Reporting more figures than the bounds support is the commonest fault in practical work.

02 Worked examples

EXAMPLE 1 A rectangle measures 4.2 m by 3.5 m, each to 1 d.p. Find the bounds of the perimeter.

① $4.15 \leq a < 4.25$ and $3.45 \leq b < 3.55$

Half of 0.1 either way.

② $P = 2(a + b)$

③ $LB = 2(4.15 + 3.45) = 15.2$

Both lower bounds.

④ $UB = 2(4.25 + 3.55) = 15.6$

Both upper bounds.

ANSWER

$$15.2 \text{ m} \leq P < 15.6 \text{ m}$$

EXAMPLE 2 The same rectangle: find the bounds of the area.

① $LB = 4.15 \times 3.45 = 14.3175$

Both lower bounds.

② $UB = 4.25 \times 3.55 = 15.0875$

Both upper bounds.

③ The bounds disagree already in the units digit.

④ So the area is “about 15 m²” — one significant figure is all that is justified.

ANSWER

$$14.32 \text{ m}^2 \leq S < 15.09 \text{ m}^2, \text{ so } \approx 15 \text{ m}^2$$

EXAMPLE**3**

A car travels 120 km (to the nearest km) in 2.0 hours (to 1 d.p.). Find the greatest possible average speed.

① $119.5 \leq d < 120.5$

Distance bounds.

② $1.95 \leq t < 2.05$

Time bounds.

③ For the greatest speed: most distance, least time.

A quotient — opposite bounds.

④ $v < \frac{120.5}{1.95} \approx 61.8 \text{ km/h}$

ANSWER

$\approx 61.8 \text{ km/h}$

03 Interactive model

Ask for the two extreme cases before any arithmetic — the rule then writes itself.

Which bound goes with which?

$a = 7 \pm 0.5, b = 4 \pm 0.5$. Largest $a + b$?

- ① A sum grows when either part grows.

So
both
go
up.

- ② $UB(a) = 7.5$

, $UB(b) = 4.5$

- ③ $a + b < 12$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Upper bound	Yuqori chegara	Верхняя граница
Lower bound	Quyi chegara	Нижняя граница
Rounded to	Yaxlitlangan	Округлено до
Nearest	Eng yaqin	Ближайший
Half a unit	Birlikning yarmi	Половина единицы
Sum	Yig'indi	Сумма
Difference	Ayirma	Разность
Product	Ko'paytma	Произведение
Quotient	Bo'linma	Частное

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

7 cm to the nearest cm has bounds:

6.9 and 7.1

6.5 and 7.5

6 and 8

6.95 and 7.05

For the largest value of $a - b$ you take:

both upper bounds

both lower bounds

UB(a) and LB(b)

LB(a) and UB(b)

For the largest value of $a \div b$ you take:

UB(a) $\div$ UB(b)

UB(a) $\div$ LB(b)

LB(a) $\div$ LB(b)

LB(a) $\div$ UB(b)

250 g to the nearest 10 g has lower bound:

249.5

245

240

249

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *9 cm to the nearest cm. Give the bounds.*

ANSWER $8.5 \leq \ell < 9.5$

2. *3.2 m to 1 d.p. Give the bounds.*

ANSWER $3.15 \leq \ell < 3.25$

3. *80 kg to the nearest 10 kg. Give the bounds.*

ANSWER $75 \leq m < 85$

4. *2.50 s to 2 d.p. Give the bounds.*

ANSWER $2.495 \leq t < 2.505$

5. *$a = 5 \pm 0.5$, $b = 3 \pm 0.5$. Largest $a + b$.*

ANSWER 9

6. *Same values: smallest $a + b$.*

ANSWER 7

7. *600 m to the nearest 100 m. Give the bounds.*

ANSWER $550 \leq d < 650$

Medium · the standard question

7 PROBLEMS

1. $a = 7 \pm 0.5$, $b = 4 \pm 0.5$. Largest $a - b$.

ANSWER 4

2. Same values: smallest $a - b$.

ANSWER 2

3. A rectangle 4.2 m by 3.5 m (1 d.p.). Bounds of the perimeter.

ANSWER $15.2 \leq P < 15.6$

4. The same rectangle: bounds of the area.

ANSWER $14.3175 \leq S < 15.0875$

5. $d = 120 \text{ km} \pm 0.5$, $t = 2.0 \text{ h} \pm 0.05$. Largest speed.

ANSWER $\approx 61.8 \text{ km/h}$

6. The same values: smallest speed.

ANSWER $\approx 58.3 \text{ km/h}$

7. A square has side 8 cm to the nearest cm. Bounds of the area.

ANSWER $56.25 \leq S < 72.25$

Hard · stretch and reason

7 PROBLEMS

1. A cube has edge 5 cm to the nearest cm. Bounds of its volume.

ANSWER $91.125 \leq V < 166.375$

2. To how many significant figures may that volume be given?

ANSWER None safely — the bounds do not even share the first digit; report “between 91 and 167 cm³”.

3. $a = 12.4 \pm 0.05$, $b = 3.1 \pm 0.05$. Bounds of $\frac{a}{b}$

ANSWER $\frac{12.35}{3.15} \approx 3.92$ to $\frac{12.45}{3.05} \approx 4.08$

4. A triangle has base 9 cm and height 6 cm, both to the nearest cm. Bounds of its area.

ANSWER $\frac{1}{2} \cdot 8.5 \cdot 5.5 = 23.375$ to $\frac{1}{2} \cdot 9.5 \cdot 6.5 = 30.875$

5. A runner covers 400 m (nearest m) in 52 s (nearest s). Bounds of the average speed.

ANSWER $\frac{399.5}{52.5} \approx 7.61$ to $\frac{400.5}{51.5} \approx 7.78$ m/s

6. Explain why the upper bound uses $<$ and the lower bound uses $\leq$.

ANSWER A value exactly on the lower bound rounds up to the stated figure; one exactly on the upper bound rounds up to the next figure instead.

7. Two lengths are 5.0 cm and 5.00 cm. Which is stated more precisely, and why?

ANSWER 5.00 cm — it claims the nearest 0.01, so bounds 4.995 to 5.005, ten times tighter.

07 Homework

Homework — 6 problems

Algebra 8, pp. 118–121, and Cambridge Learner’s Book 3.4.

- 15 cm to the nearest cm. Give the bounds.
- 6.70 kg to 2 d.p. Give the bounds.
- $a = 20 \pm 0.5$, $b = 8 \pm 0.5$. Find the bounds of $a - b$.

4. *A rectangle is 7.5 cm by 4.2 cm, both to 1 d.p. Find the bounds of the area.*
5. *A car covers 250 km (nearest km) in 4.0 h (1 d.p.). Find the bounds of the average speed.*
6. *A square has side 12 cm to the nearest cm. Find the bounds of the perimeter.*

Control work 5 · Inequalities, modulus and accuracy

The Chapter II assessment, then a work-on-mistakes lesson sorted by error type.

LESSON 60–61

2 × 40 MIN

ALGEBRA 8, §§15–19

STAGE 9 · 3.4, 4.3

LESSON CLOCK

4'

72'

4'

Setting up	4 min
The paper	72 min
Collect in	4 min

LEARNING OBJECTIVES

- Assess inequalities in one unknown, systems, modulus, error and bounds.
- Sort every lost mark into one of four error types.
- Re-solve every task that was lost.

TEXTBOOK

National	Revision of §§15–19
Cambridge	Learner’s Book pp. 74–81, 96–102
Workbook	Workbook 3.4, 4.3

01

Explanation

Lesson 60 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each:

- tasks 1–2 · solve a linear inequality, one with a negative coefficient (§15)
- task 3 · solve a system and write the interval (§16)
- task 4 · a double inequality (§16)
- task 5 · a modulus equation (§17)
- task 6 · a modulus inequality (§17)
- task 7 · bounds or relative error (§§18–19)

Lesson 61 — work on mistakes (40 minutes)

THE FOUR ERRORS THIS PAPER PRODUCES

- The sign not reversed after dividing by a negative number.
- Only one case taken for a modulus equation — the negative case forgotten.
- The two modulus inequality shapes confused — writing two rays where one interval belongs, or the reverse.

4. **Bracket types mixed up** when writing an interval, or ∞ given a square bracket.

02 Worked examples

EXAMPLE 1 Find the error: $|x - 2| = 5$, so $x = 7$

- ① Only the positive case was taken.
This is error 2.

- ② $x - 2 = 5$ or $x - 2 = -5$
Both cases.

- ③ $x = 7$ or $x = -3$

ANSWER

$$x = 7 \text{ or } x = -3$$

EXAMPLE 2 Find the error: $|x| > 3$, so $-3 < x < 3$

- ① That is the answer for $|x| < 3$, not for $>$.
The two shapes are swapped.

- ② “More than 3 from zero” is everything **outside**.

- ③ $x < -3$ or $x > 3$

ANSWER

$$(-\infty; -3) \cup (3; +\infty)$$

03 Interactive model

Show the wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: $-3x < 12 \Rightarrow x < -4$

① Dividing by

-3 — a negative number.

② The sign turns:

$x > -4$

③ Test

$x = 0$:

$0 < 12$ ✓, and

$0 > -4$ ✓.

One
test
settles
it.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Solution set	Yechimlar to'plami	Множество решений
Interval	Oraliq	Промежуток
Modulus	Modul	Модуль
Bound	Chegara	Граница
Error	Xato	Ошибка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$-3x < 12$ gives:

$$x < -4$$

$$x > -4$$

$$x < 4$$

$$x > 4$$

$|x - 2| = 5$ gives:

$$x = 7$$

$$x = -3$$

$$x = 7 \text{ or } x = -3$$

no solutions

$|x| > 3$ gives:

$$-3 < x < 3$$

$$x < -3 \text{ or } x > 3$$

$$x > 3$$

$$x < 3$$

$x \geq 2$ as an interval is:

$(2; +\infty)$

$[2; +\infty)$

$(2; +\infty]$

$[2; +\infty]$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** Solve $4x - 5 > 7$

ANSWER $x > 3$

2. **Task 2.** Solve $-2x + 1 \geq 9$

ANSWER $x \leq -4$

3. **Task 3.** Solve: $x > 2$ and $x \leq 6$

ANSWER $(2; 6]$

4. **Task 4.** Solve $-3 < 2x + 1 \leq 7$

ANSWER $(-2; 3]$

5. **Task 5.** Solve $|x - 1| = 4$

ANSWER $x = 5$ or $x = -3$

6. **Task 6.** Solve $|x| \leq 5$

ANSWER $[-5; 5]$

7. **Task 7.** 8 m to the nearest metre. Give the bounds.

ANSWER $7.5 \leq \ell < 8.5$

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** Solve $5x + 2 \leq 17$

ANSWER $x \leq 3$

2. **Task 2.** Solve $3 - 4x > 15$

ANSWER $x < -3$

3. **Task 3.** Solve: $x \geq -1$ and $x < 4$

ANSWER $[-1; 4)$

4. **Task 4.** Solve $-6 \leq 3x < 9$

ANSWER $[-2; 3)$

5. **Task 5.** Solve $|2x + 1| = 9$

ANSWER $x = 4$ or $x = -5$

6. **Task 6.** Solve $|x - 2| > 3$

ANSWER $x < -1$ or $x > 5$

7. **Task 7.** Exact 40, approximation 39. Find the relative error.

ANSWER 2.5%

Hard · stretch and reason

7 PROBLEMS

1. Find the error: $-3x < 12$, so $x < -4$

ANSWER The sign was not reversed. Correct: $x > -4$.

2. Find the error: $|x - 2| = 5$, so $x = 7$

ANSWER Only one case. Correct: $x = 7$ or $x = -3$.

3. Find the error: $|x| > 3$, so $-3 < x < 3$

ANSWER The shapes are swapped. Correct: $x < -3$ or $x > 3$.

4. Find the error: $x \geq 2$ written as $(2; +\infty]$

ANSWER Correct: $[2; +\infty)$.

5. Find the error: $|x + 4| = -1$, so $x = -5$ or $x = -3$

ANSWER A modulus is never negative — there are no solutions.

6. Find the error: “7 cm to the nearest cm lies between 6.9 and 7.1”

ANSWER The bounds are 6.5 and 7.5 — half a unit, not a tenth.

7. Find the error: “ $x > 5$ and $x < 2$ gives $(5; 2)$ ”

ANSWER The sets do not overlap; there are no solutions.

07 Homework

After the work-on-mistakes lesson

Re-solve every task you lost marks on. Quadratic equations begin next lesson.

- Solve $-5x + 3 \geq 18$
- Solve $|3x - 2| = 10$
- Solve $|x + 1| < 6$
- A length reads 9.4 cm to 1 d.p. Give the bounds.
- Write out the four errors from this lesson in your own words with an example each.

The quadratic equation and its roots

The first equation of the course whose graph is not a line — and the first that can have two answers, one, or none at all.

LESSON 62–63

2 × 40 MIN

ALGEBRA 8, CHAPTER III \$20

STAGE 9 · BEYOND

LESSON CLOCK

10'

24'

16'

26'

4'

Warm-up	10 min
Explanation	24 min
Interactive	16 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- Recognise a quadratic equation and name its coefficients.
- Check whether a given number is a root.
- Solve a quadratic by factorising.
- Read the number of roots off the graph.

TEXTBOOK

National	Chapter III, pp. 122–128
Cambridge	Extension beyond Stage 9
Workbook	—

01

Explanation

What a quadratic equation is

DEFINITION

An equation of the form

$$ax^2 + bx + c = 0, \quad a \neq 0$$

is a **quadratic equation**. Here a is the **leading coefficient**, b the second coefficient and c the **free term**.

The condition $a \neq 0$ matters: without it the x^2 disappears and the equation is linear.

EQUATION	A	B	C
$x^2 - 5x + 6 = 0$	1	-5	6
$3x^2 + 2x - 8 = 0$	3	2	-8
$2x^2 - 7 = 0$	2	0	-7
$x^2 + 4x = 0$	1	4	0

Write the equation in this standard order **before** reading off the coefficients — and mind the signs: in $x^2 - 5x + 6$ the coefficient b is -5 , not 5 .

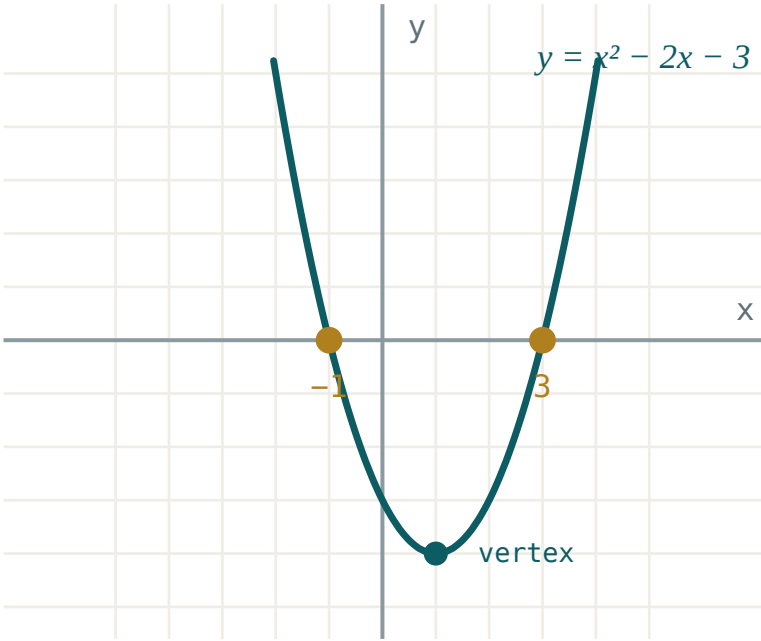
Roots

DEFINITION

A **root** of the equation is a value of x that makes it true.

To check, substitute. Is 3 a root of $x^2 - 5x + 6 = 0$? $9 - 15 + 6 = 0$ ✓ — yes.

Graphically, the roots are where the parabola $y = ax^2 + bx + c$ crosses the x -axis.



The graph of $y = x^2 - 2x - 3$ crosses the x -axis at -1 and 3 — the two roots of $x^2 - 2x - 3 = 0$.

A parabola can cross the axis twice, touch it once, or miss it entirely — so a quadratic equation has **two roots, one root, or none**. That is the single biggest difference from the linear equations of Grade 7.

Solving by factorising

THE ZERO PRODUCT RULE

If $A \cdot B = 0$ then $A = 0$ or $B = 0$. Nothing else can happen — this is the only reason factorising solves equations.

So write the quadratic as a product equal to zero, then set each bracket to zero in turn:

$$x^2 - 5x + 6 = 0$$

$$(x - 2)(x - 3) = 0$$

$$x - 2 = 0 \text{ or } x - 3 = 0 \implies x = 2 \text{ or } x = 3$$

To factorise $x^2 + px + q$, find two numbers with **product** q and **sum** p . For $x^2 - 5x + 6$: product 6, sum $-5 \rightarrow -2$ and -3 .

THE RULE NEEDS A ZERO ON THE RIGHT

From $(x - 1)(x - 4) = 6$ you may **not** write $x - 1 = 6$. Expand, move everything to one side, and factorise again.

02 Worked examples

EXAMPLE 1 Solve $x^2 - 7x + 12 = 0$

- 1 Two numbers with product 12 and sum -7 .
Both negative, since the product is positive and the sum negative.
- 2 -3 and -4
- 3 $(x - 3)(x - 4) = 0$
- 4 $x = 3$ or $x = 4$
Check: $9 - 21 + 12 = 0$ ✓

ANSWER

$$x = 3, x = 4$$

EXAMPLE 2 Solve $x^2 + 2x - 15 = 0$

- 1 Product -15 , sum 2 .
Opposite signs, since the product is negative.
- 2 5 and -3
- 3 $(x + 5)(x - 3) = 0$
- 4 $x = -5$ or $x = 3$

ANSWER

$$x = -5, x = 3$$

EXAMPLE 3 *Is $x = -2$ a root of $2x^2 + 3x - 2 = 0$?*

① $2(-2)^2 + 3(-2) - 2$
Substitute.

② $= 8 - 6 - 2$
Mind the sign: $(-2)^2 = 4$, not -4 .

③ $= 0$
It works.

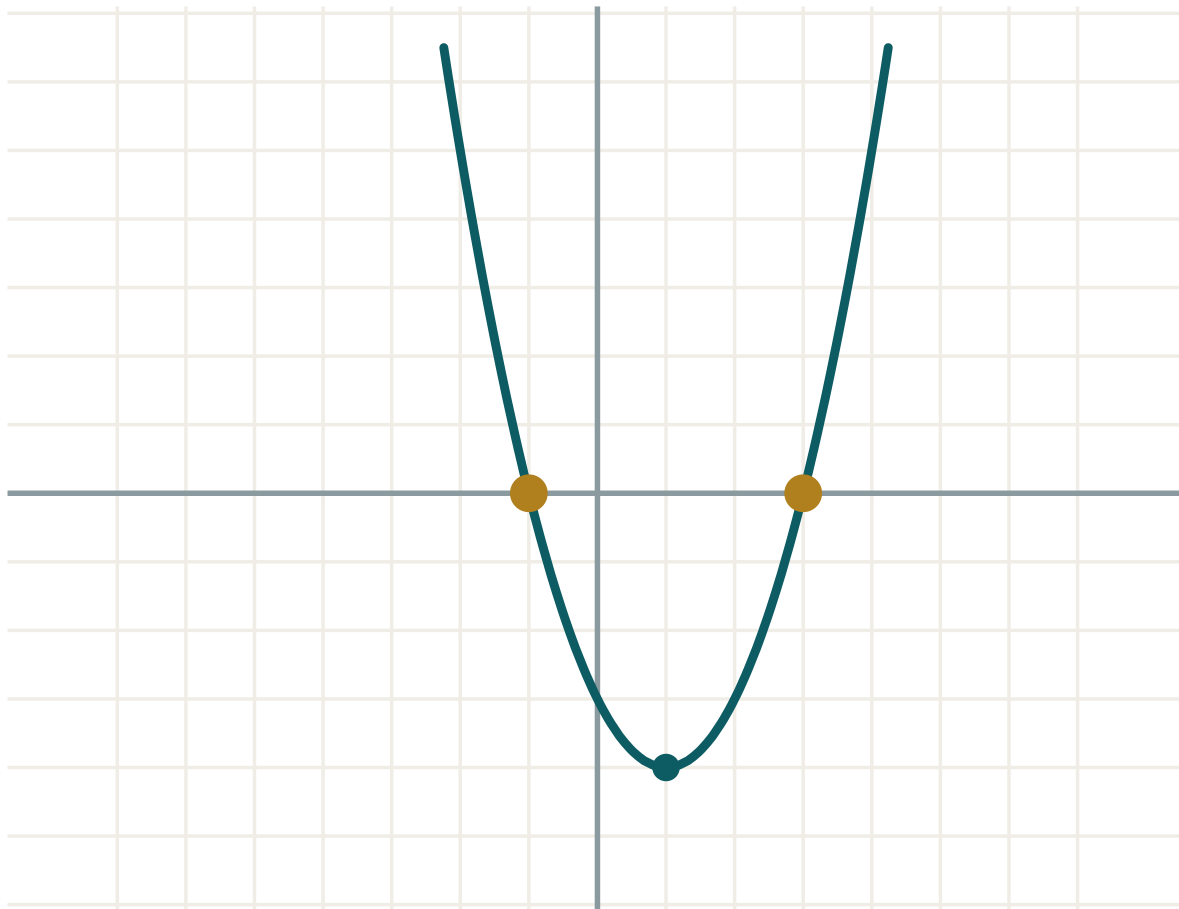
ANSWER

Yes, $x = -2$ is a root.

03 Interactive model

Set $a = 1$, $b = -2$, $c = -3$ and read the roots off the graph before doing any algebra.

Roots and the discriminant

a **1**b **-2**c **-3** $D = b^2 - 4ac$ **16**roots **two** x_1 **-1** x_2 **3** $x_1 + x_2$ **2** $x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Quadratic equation	Kvadrat tenglama	Квадратное уравнение
Root of an equation	Tenglamaning ildizi	Корень уравнения
Coefficient	Koeffitsiyent	Коэффициент
Leading coefficient	Bosh koeffitsiyent	Старший коэффициент
Free term (constant)	Ozod had	Свободный член
Factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Zero product rule	Ko'paytmaning nolga tengligi	Произведение равно нулю
Parabola	Parabola	Парабола
Substitute and check	O'rniga qo'yib tekshirish	Подставить и проверить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In $3x^2 - 5x + 2 = 0$, the coefficient b is:

A quadratic equation can have:

$(x - 2)(x + 5) = 0$ gives:

$x^2 - 5x + 6 = 0$ factorises to:

$$(x - 2)(x - 3)$$

$$(x + 2)(x + 3)$$

$$(x - 1)(x - 6)$$

$$(x - 2)(x + 3)$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Name a, b, c in $x^2 + 3x - 4 = 0$

ANSWER $1, 3, -4$

2. Name a, b, c in $2x^2 - 5 = 0$

ANSWER $2, 0, -5$

3. Is $x = 2$ a root of $x^2 - 4 = 0$?

ANSWER Yes.

4. Is $x = 1$ a root of $x^2 + x - 2 = 0$?

ANSWER Yes.

5. Solve $(x - 3)(x + 1) = 0$

ANSWER $x = 3, x = -1$

6. Solve $(x + 4)(x + 2) = 0$

ANSWER $x = -4, x = -2$

7. Solve $x(x - 5) = 0$

ANSWER $x = 0, x = 5$

Medium · the standard question

7 PROBLEMS

1. Solve $x^2 - 7x + 12 = 0$

ANSWER $x = 3, x = 4$

2. Solve $x^2 + 2x - 15 = 0$

ANSWER $x = -5, x = 3$

3. Solve $x^2 - x - 6 = 0$

ANSWER $x = 3, x = -2$

4. Solve $x^2 + 7x + 10 = 0$

ANSWER $x = -2, x = -5$

5. Solve $x^2 - 9x + 20 = 0$

ANSWER $x = 4, x = 5$

6. Is $x = -2$ a root of $2x^2 + 3x - 2 = 0$?

ANSWER Yes.

7. Solve $x^2 - 2x - 8 = 0$

ANSWER $x = 4, x = -2$

Hard · stretch and reason

7 PROBLEMS

1. Solve $2x^2 - 5x + 3 = 0$ by factorising

ANSWER $(2x - 3)(x - 1) = 0$, so $x = 1.5, x = 1$

2. Solve $3x^2 + 5x - 2 = 0$

ANSWER $(3x - 1)(x + 2) = 0$, so $x = \frac{1}{3}, x = -2$

3. Solve $(x - 1)(x - 4) = 6$

ANSWER $x^2 - 5x - 2 = 0$ — does not factorise nicely; the roots come from the formula next lesson.

4. One root of $x^2 + bx - 6 = 0$ is 2. Find b and the other root.

ANSWER $4 + 2b - 6 = 0$ gives $b = 1$; the other root is -3 .

5. For which c does $x^2 - 6x + c = 0$ have $x = 2$ as a root?

ANSWER $4 - 12 + c = 0$, so $c = 8$

6. Solve $x^2 = 5x$

ANSWER $x(x - 5) = 0$: $x = 0$ or $x = 5$. Never divide by x — you would lose the root 0.

7. Write a quadratic equation whose roots are 3 and -7 .

ANSWER $(x - 3)(x + 7) = 0$, i.e. $x^2 + 4x - 21 = 0$

07 Homework

Homework — 6 problems

Algebra 8, Chapter III, pp. 122–128. Check every root by substituting.

1. Name a, b, c in $4x^2 - 7x + 1 = 0$
2. Solve $x^2 - 8x + 15 = 0$
3. Solve $x^2 + 5x - 14 = 0$
4. Solve $x^2 = 7x$

5. Is $x = -3$ a root of $x^2 + x - 6 = 0$?
6. Write a quadratic equation with roots 2 and -5

Incomplete quadratic equations

When b or c is zero the formula is unnecessary — three short methods that are quicker and safer.

LESSON 64–65

2 × 40 MIN

ALGEBRA 8, CHAPTER III §21

STAGE 9 · BEYOND

LESSON CLOCK

10'

24'

16'

26'

4'

Warm-up	10 min
Explanation	24 min
Interactive	16 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- Recognise the three incomplete forms.
- Solve $ax^2 + c = 0$ by taking roots.
- Solve $ax^2 + bx = 0$ by taking out a common factor.
- Explain why you must never divide an equation by x .

TEXTBOOK

National	Chapter III, pp. 129–133
Cambridge	Extension beyond Stage 9
Workbook	–

01

Explanation

The three forms

FORM	MISSING	METHOD	ROOTS
$ax^2 = 0$	b and c	divide by a	$x = 0$ only
$ax^2 + bx = 0$	c	common factor x	$x = 0$ and $x = -\frac{b}{a}$
$ax^2 + c = 0$	b	make x^2 the subject	two, or none

Form 1 — no b

$$ax^2 + c = 0 \implies x^2 = -\frac{c}{a}$$

- If $-\frac{c}{a} > 0$ there are **two roots**, $x = \pm\sqrt{-c/a}$.
- If it is 0, one root, $x = 0$.
- If it is **negative**, there are **no roots** — no real number squares to a negative.

$$2x^2 - 18 = 0 \implies x^2 = 9 \implies x = \pm 3$$

$$x^2 + 4 = 0 \implies x^2 = -4 \implies \text{no roots}$$

DO NOT LOSE THE MINUS ROOT

$x^2 = 9$ has **two** answers, 3 and -3 . The symbol $\sqrt{9}$ means only 3 — that is why the $\pm$ has to be written by hand.

Form 2 — no c

$$ax^2 + bx = 0 \implies x(ax + b) = 0 \implies x = 0 \text{ or } x = -\frac{b}{a}$$

$$3x^2 - 12x = 0 \implies 3x(x - 4) = 0 \implies x = 0 \text{ or } x = 4$$

NEVER DIVIDE BY X

Dividing $3x^2 = 12x$ by x gives $3x = 12$, so $x = 4$ — and the root $x = 0$ has vanished. You may only divide by something you know is not zero, and x might be. Always move everything to one side and factorise instead.

An equation of this form **always** has $x = 0$ as a root, because the free term is zero.

02 Worked examples**EXAMPLE 1** Solve $5x^2 - 45 = 0$

① $5x^2 = 45$

② $x^2 = 9$
Divide by 5.

③ $x = 3$ or $x = -3$
Both square roots.

ANSWER

$$x = \pm 3$$

EXAMPLE 2 Solve $4x^2 + 7x = 0$

① $x(4x + 7) = 0$
Common factor x .

② $x = 0$ or $4x + 7 = 0$
Zero product rule.

③ $x = 0$ or $x = -1.75$

ANSWER

$$x = 0, x = -1.75$$

EXAMPLE 3 Solve $3x^2 + 12 = 0$

① $3x^2 = -12$

② $x^2 = -4$

③ A square is never negative.

④ No real roots.

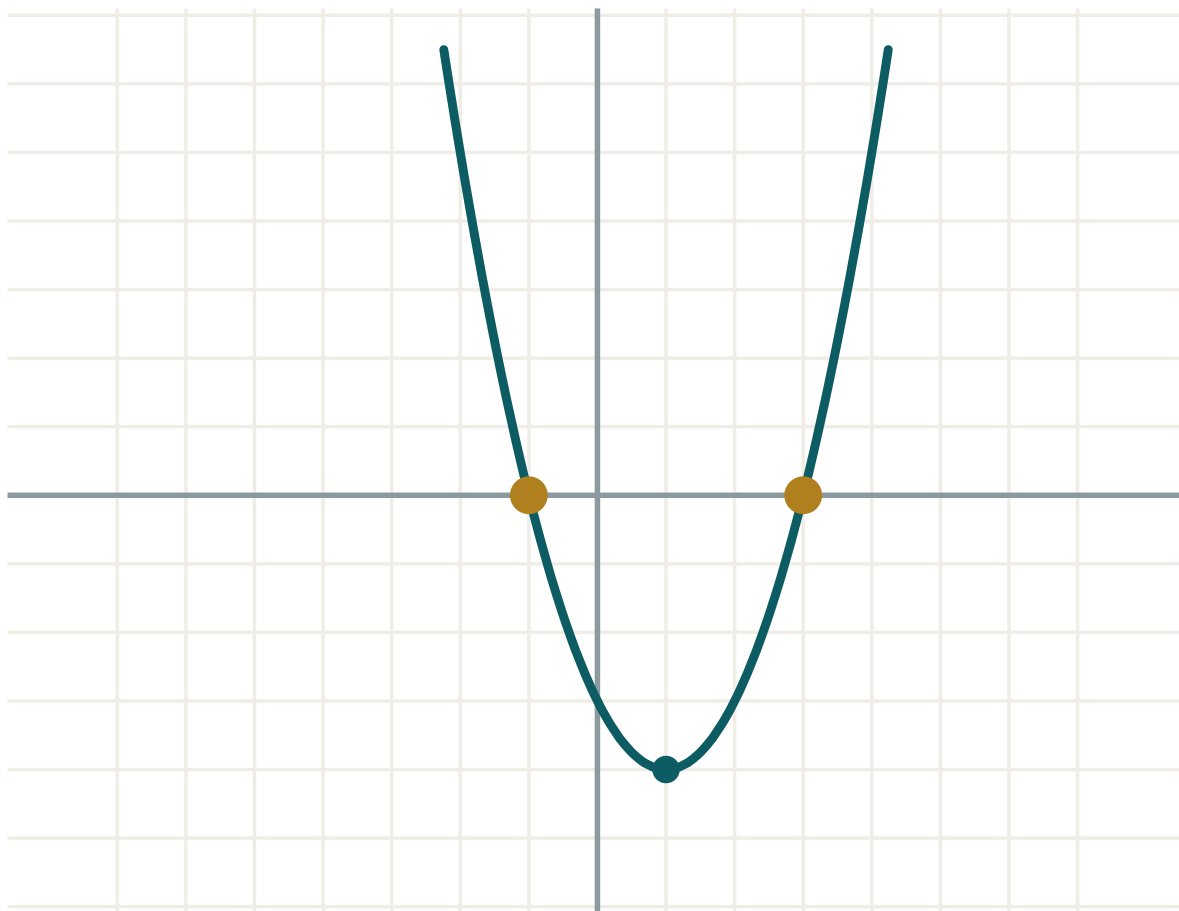
ANSWER

No roots.

03 Interactive model

Set $b = 0$ and vary c , then set $c = 0$ and vary b — the two incomplete shapes appear on the graph.

Set b or c to zero



a 1

b -2

c -3

 $D = b^2 - 4ac$ 16

roots two

 x_1 -1 x_2 3 $x_1 + x_2$ 2 $x_1 \cdot x_2$ -3

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Incomplete quadratic equation	Chala kvadrat tenglama	Неполное квадратное уравнение
Common factor	Umumiy ko'paytuvchi	Общий множитель
Take the square root	Kvadrat ildiz chiqarish	Извлечь квадратный корень
Two opposite roots	Qarama-qarshi ildizlar	Противоположные корни
Lose a root	Ildizni yo'qotish	Потерять корень
No real roots	Haqiqiy ildizga ega emas	Нет действительных корней
Zero root	Nol ildiz	Нулевой корень

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself** $x^2 = 16$ gives:

$x = 4$

$x = -4$

$x = \pm 4$

$x = 8$

 $x^2 + 9 = 0$ has:

$x = \pm 3$

$x = 3$

no roots

$x = 0$

 $x^2 - 6x = 0$ gives:

$x = 6$

$x = 0, x = 6$

$x = \pm 6$

$x = 0$

Dividing $x^2 = 5x$ by x is wrong because:

it is too slow

it loses the root $x = 0$

x^2 is not divisible by x

the answer becomes negative

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $x^2 = 25$

ANSWER $x = \pm 5$

2. Solve $x^2 - 49 = 0$

ANSWER $x = \pm 7$

3. Solve $x^2 + 1 = 0$

ANSWER No roots.

4. Solve $x^2 = 0$

ANSWER $x = 0$

5. Solve $x^2 - 3x = 0$

ANSWER $x = 0, x = 3$

6. Solve $x^2 + 5x = 0$

ANSWER $x = 0, x = -5$

7. Solve $2x^2 = 8$

ANSWER $x = \pm 2$

Medium · the standard question

7 PROBLEMS

1. Solve $5x^2 - 45 = 0$

ANSWER $x = \pm 3$

2. Solve $4x^2 + 7x = 0$

ANSWER $x = 0, x = -1.75$

3. Solve $3x^2 + 12 = 0$

ANSWER No roots.

4. Solve $3x^2 - 12x = 0$

ANSWER $x = 0, x = 4$

5. Solve $9x^2 - 4 = 0$

ANSWER $x = \pm \frac{2}{3}$

6. Solve $x^2 = 7x$

ANSWER $x = 0, x = 7$

7. Solve $2x^2 - 50 = 0$

ANSWER $x = \pm 5$

Hard · stretch and reason

7 PROBLEMS

1. Solve $(x - 2)^2 = 9$

ANSWER $x - 2 = \pm 3$: $x = 5$ or $x = -1$

2. Solve $4x^2 = 3x$

ANSWER $x(4x - 3) = 0$: $x = 0$ or $x = 0.75$

3. For which c does $x^2 + c = 0$ have two roots?

ANSWER $c < 0$

4. Solve $2x^2 - 6 = 0$ and give the answer exactly.

ANSWER $x^2 = 3$, so $x = \pm\sqrt{3}$

5. Solve $x(x + 3) = 4x$

ANSWER $x^2 + 3x - 4x = 0 \implies x(x - 1) = 0$: $x = 0$, $x = 1$

6. The area of a square is 121 cm^2 . Find its side, and say why only one answer makes sense.

ANSWER $x^2 = 121$ gives $x = \pm 11$; a length cannot be negative, so 11 cm .

7. Solve $(2x - 1)^2 = 0$

ANSWER $2x - 1 = 0$, so $x = 0.5$ — one root only.

07 Homework

Homework — 6 problems

Algebra 8, Chapter III, pp. 129–133. Never divide by x .

1. Solve $x^2 - 64 = 0$

2. Solve $3x^2 - 27 = 0$

3. Solve $x^2 + 16 = 0$

4. Solve $5x^2 + 20x = 0$

5. Solve $x^2 = 9x$

6. Solve $(x + 1)^2 = 16$

The quadratic formula and the discriminant

One formula that solves every quadratic — and one number inside it that says in advance how many roots there will be.

LESSON 66–67

2 × 40 MIN

ALGEBRA 8, CHAPTER III §22

STAGE 9 · BEYOND

LESSON CLOCK

10'

24'

16'

26'

4'

Warm-up	10 min
Explanation	24 min
Interactive	16 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- Calculate the discriminant $D = b^2 - 4ac$.
- Say how many roots an equation has from the sign of D .
- Use the quadratic formula correctly, including with negative coefficients.
- Choose between factorising and the formula.

TEXTBOOK

National	Chapter III, pp. 134–140
Cambridge	Extension beyond Stage 9
Workbook	—

01

Explanation

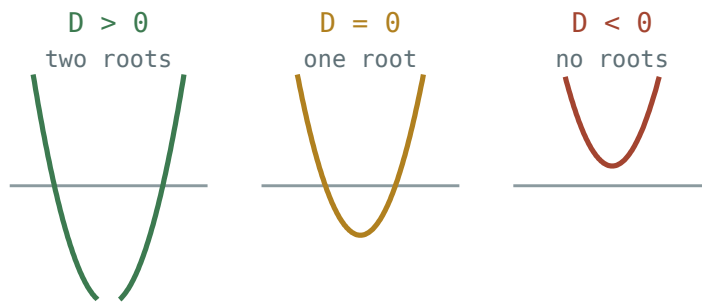
The discriminant

DEFINITION

For $ax^2 + bx + c = 0$ the **discriminant** is

$$D = b^2 - 4ac$$

D	ROOTS	THE PARABOLA
$D > 0$	two different roots	crosses the x-axis twice
$D = 0$	one root (repeated)	touches the axis
$D < 0$	no real roots	misses the axis



The sign of the discriminant decides whether the parabola crosses, touches or misses the x-axis.

Work out D **first**, every time. If it is negative you can stop — there is nothing to find, and you have saved yourself a page of arithmetic.

The formula

THE QUADRATIC FORMULA

$$x = \frac{-b \pm \sqrt{D}}{2a}, \text{ where } D = b^2 - 4ac$$

It comes from completing the square on the general equation, and it works for every quadratic whatever the coefficients.

THREE PLACES WHERE MARKS ARE LOST

1. $-b$ when b is already negative: for $b = -5$, $-b = +5$.
2. $-4ac$ when c is negative: $-4 \cdot 1 \cdot (-3) = +12$.
3. The whole of $-b \pm \sqrt{D}$ is divided by $2a$ — not just the root.

Which method?

- **Incomplete** ($b = 0$ or $c = 0$) → the short methods of the last lesson.
- **Factorises easily** (small whole-number roots) → factorise; it is quicker.
- **Anything else** → the formula.

A useful check: if D is a perfect square, the equation would have factorised. If it is not, the roots are irrational and the formula was the only route.

02 Worked examples

EXAMPLE 1 Solve $2x^2 - 7x + 3 = 0$

① $a = 2, b = -7, c = 3$

Read off the coefficients with their signs.

② $D = (-7)^2 - 4 \cdot 2 \cdot 3 = 49 - 24 = 25$

$D > 0$, so two roots.

③ $x = \frac{7 \pm \sqrt{25}}{4} = \frac{7 \pm 5}{4}$

$-b = +7$

④ $x = 3$ or $x = 0.5$

$\frac{12}{4}$ and $\frac{2}{4}$

ANSWER

$x = 3, x = 0.5$

EXAMPLE 2 Solve $x^2 + 4x + 4 = 0$

① $a = 1, b = 4, c = 4$

② $D = 16 - 16 = 0$

One repeated root.

③ $x = \frac{-4}{2} = -2$

④ $(x + 2)^2 = 0$

The same answer by factorising.

ANSWER

$x = -2$

EXAMPLE 3 Solve $x^2 - 2x + 5 = 0$

① $a = 1, b = -2, c = 5$

② $D = 4 - 20 = -16$

$D < 0.$

③ No real roots — stop here.

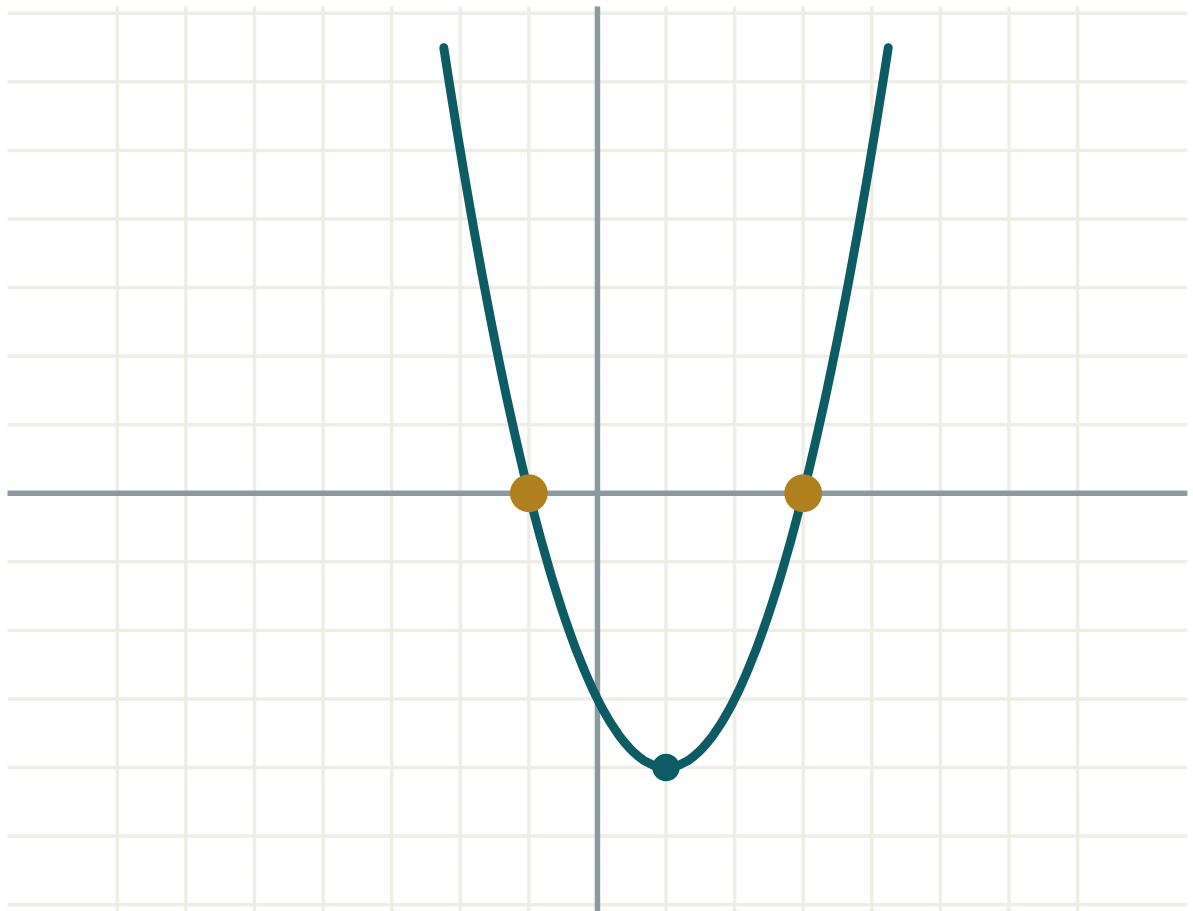
The parabola lies entirely above the x-axis.

ANSWER

No real roots.

03 Interactive model

Move c downwards and watch D pass through zero as the parabola touches then crosses the axis.

D decides everything**a 1****b -2****c -3** **$D = b^2 - 4ac$ 16****roots two** **x_1 -1** **x_2 3** **$x_1 + x_2$ 2** **$x_1 \cdot x_2$ -3****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Discriminant	Diskriminant	Дискриминант
Quadratic formula	Kvadrat tenglama formulasi	Формула корней квадратного уравнения
Two distinct roots	Ikkita har xil ildiz	Два различных корня
Repeated root	Karrali ildiz	Кратный корень
No real roots	Haqiqiy ildiz yo'q	Нет действительных корней
Substitute	O'rniga qo'yish	Подставить
Simplify the surd	Ildizni soddalashtirish	Упростить корень
Exact answer	Aniq javob	Точный ответ

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

For $x^2 - 5x + 6 = 0$, D is:

$D = 0$ means:

For $2x^2 + 3x + 5 = 0$, D is:

In the formula, when $b = -6$ the numerator starts with:

-6

$+6$

6^2

-36

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find D for $x^2 + 3x + 2 = 0$

ANSWER 1

2. Find D for $x^2 - 4x + 4 = 0$

ANSWER 0

3. Find D for $x^2 + x + 1 = 0$

ANSWER -3

4. How many roots when $D = 9$?

ANSWER Two.

5. How many roots when $D = 0$?

ANSWER One.

6. How many roots when $D = -5$?

ANSWER None.

7. Solve $x^2 + 3x + 2 = 0$

ANSWER $x = -1, x = -2$

Medium · the standard question

7 PROBLEMS

1. Solve $2x^2 - 7x + 3 = 0$

ANSWER $x = 3, x = 0.5$

2. Solve $x^2 + 4x + 4 = 0$

ANSWER $x = -2$

3. Solve $x^2 - 2x + 5 = 0$

ANSWER No real roots.

4. Solve $3x^2 - 5x - 2 = 0$

ANSWER $D = 49: x = 2, x = -\frac{1}{3}$

5. Solve $x^2 - 6x + 9 = 0$

ANSWER $x = 3$

6. Solve $2x^2 + 5x - 3 = 0$

ANSWER $D = 49: x = 0.5, x = -3$

7. Solve $x^2 + 2x - 8 = 0$

ANSWER $x = 2, x = -4$

Hard · stretch and reason

7 PROBLEMS

1. Solve $x^2 - 4x + 1 = 0$ exactly

ANSWER $D = 12$: $x = 2 \pm \sqrt{3}$

2. Solve $2x^2 - 4x - 1 = 0$ exactly

ANSWER $D = 24$: $x = \frac{2 \pm \sqrt{6}}{2}$

3. For which k does $x^2 + kx + 9 = 0$ have exactly one root?

ANSWER $k^2 = 36$, so $k = \pm 6$

4. For which k does $x^2 - 6x + k = 0$ have two roots?

ANSWER $36 - 4k > 0$, so $k < 9$

5. For which k does $x^2 + 4x + k = 0$ have no roots?

ANSWER $16 - 4k < 0$, so $k > 4$

6. Solve $5x^2 - 2x - 3 = 0$

ANSWER $D = 64$: $x = 1, x = -0.6$

7. Show that $x^2 + x + 1 = 0$ has no real roots, without the formula.

ANSWER $x^2 + x + 1 = (x + 0.5)^2 + 0.75 \geq 0.75 > 0$ for every x .

07 Homework

Homework — 6 problems

Algebra 8, Chapter III, pp. 134–140. Find D before anything else.

- Find D and solve $x^2 - 7x + 10 = 0$
- Find D and solve $3x^2 + 2x - 1 = 0$
- Find D and solve $x^2 + 2x + 5 = 0$
- Solve $x^2 - 8x + 16 = 0$
- Solve $x^2 - 6x + 2 = 0$ exactly

6. For which k does $x^2 + kx + 4 = 0$ have exactly one root?

Vieta's theorem and factorising a quadratic trinomial

The sum and the product of the roots, read straight off the coefficients — a check, a shortcut and a factorising method in one.

LESSON 68–69

2 × 40 MIN

ALGEBRA 8, CHAPTER III §23

STAGE 9 · 2.4

LESSON CLOCK



Warm-up	10 min
Explanation	24 min
Interactive	16 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- State and use Vieta's theorem.
- Check a pair of roots without substituting.
- Find roots by inspection for a reduced quadratic.
- Factorise $ax^2 + bx + c$ using its roots.

TEXTBOOK

National	Chapter III, pp. 141–146
Cambridge	Learner's Book pp. 36–39
Workbook	Workbook 2.4

01 Explanation

The theorem

VIETA'S THEOREM

If x_1 and x_2 are the roots of $ax^2 + bx + c = 0$, then

$$x_1 + x_2 = -\frac{b}{a} \quad \text{and} \quad x_1 \cdot x_2 = \frac{c}{a}$$

For a **reduced** equation $x^2 + px + q = 0$ (with $a = 1$) it is even simpler:

$$x_1 + x_2 = -p \quad \text{and} \quad x_1 \cdot x_2 = q$$

Check it on $x^2 - 5x + 6 = 0$, whose roots are 2 and 3: $2 + 3 = 5 = -(-5) \checkmark$ and $2 \cdot 3 = 6 \checkmark$.

Three uses

1 · As a check. After solving, add and multiply your two roots. If they do not give $-\frac{b}{a}$ and $\frac{c}{a}$, go back — this takes five seconds and catches almost every arithmetic slip.

2 · To find roots by inspection. For $x^2 - 7x + 12 = 0$, look for two numbers with sum 7 and product 12: 3 and 4. No formula needed.

3 · To find an unknown coefficient. If one root of $x^2 + bx - 10 = 0$ is 5, then the product of the roots is -10 , so the other root is -2 , and the sum $5 + (-2) = 3 = -b$ gives $b = -3$.

Factorising the trinomial

THEOREM

If x_1 and x_2 are the roots of $ax^2 + bx + c = 0$, then

$$ax^2 + bx + c = a(x - x_1)(x - x_2)$$

So factorising a quadratic trinomial and solving the equation are the same job. Find the roots, then write the brackets:

$$2x^2 - 7x + 3: \text{ roots } 3 \text{ and } 0.5, \text{ so } = 2(x - 3)(x - 0.5) = (x - 3)(2x - 1)$$

DO NOT FORGET THE A

$2x^2 - 7x + 3$ is not $(x - 3)(x - 0.5)$ — that expands to $x^2 - 3.5x + 1.5$. The leading coefficient must be carried through.

If $D < 0$ there are no roots, and the trinomial **cannot** be factorised over the real numbers.

02 Worked examples

EXAMPLE 1 Solve $x^2 - 9x + 20 = 0$ by inspection.

- 1 Sum of roots = 9, product = 20.
Vieta for a reduced equation.
- 2 Two numbers with product 20: 1·20, 2·10, 4·5.
- 3 $4 + 5 = 9$ ✓
- 4 $x = 4$ and $x = 5$

ANSWER

$$x = 4, x = 5$$

EXAMPLE 2 One root of $x^2 + bx - 21 = 0$ is 7. Find the other root and b .

- 1 $x_1 \cdot x_2 = -21$
The product equals the free term.
- 2 $7 \cdot x_2 = -21$, so $x_2 = -3$
- 3 $x_1 + x_2 = 7 + (-3) = 4$
- 4 $-b = 4$, so $b = -4$

ANSWER

$$x_2 = -3, b = -4$$

EXAMPLE 3 *Factorise $3x^2 - 5x - 2$*

$$① \quad D = 25 + 24 = 49$$

$$② \quad x = \frac{5 \pm 7}{6}$$

The formula.

$$③ \quad x_1 = 2, x_2 = -\frac{1}{3}$$

$$④ \quad 3(x - 2)\left(x + \frac{1}{3}\right) = (x - 2)(3x + 1)$$

Multiply the 3 into the second bracket.

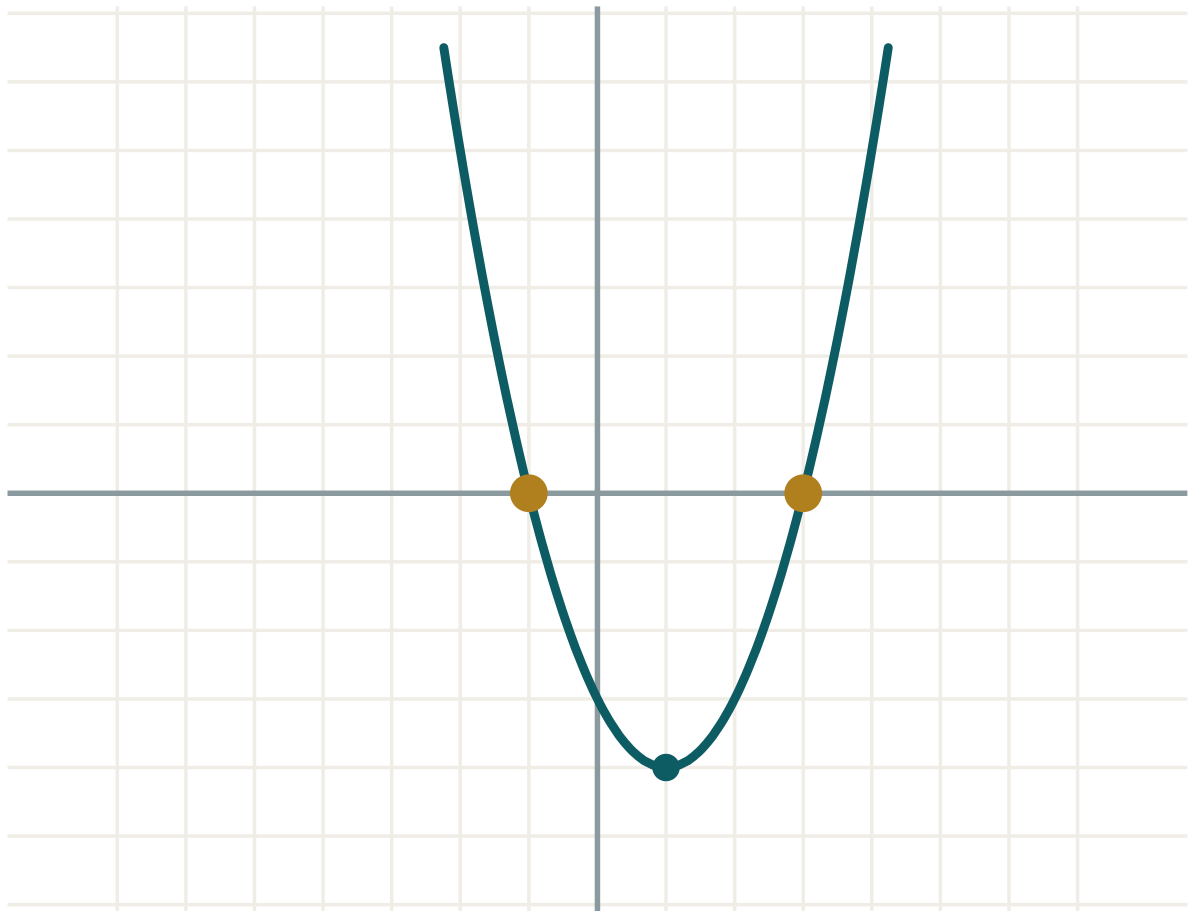
ANSWER

$$(x - 2)(3x + 1)$$

03 **Interactive model**

Read $x_1 + x_2$ and $x_1 \cdot x_2$ from the model and compare with $-b/a$ and c/a .

Sum and product of the roots



a 1

b -2

c -3

 $D = b^2 - 4ac$ 16

roots two

 x_1 -1 x_2 3 $x_1 + x_2$ 2 $x_1 \cdot x_2$ -3

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Vieta's theorem	Viyet teoremasi	Теорема Виета
Sum of the roots	Ildizlar yig'indisi	Сумма корней
Product of the roots	Ildizlar ko'paytmasi	Произведение корней
Reduced quadratic equation	Keltirilgan kvadrat tenglama	Приведённое квадратное уравнение
Quadratic trinomial	Kvadrat uchhad	Квадратный трёхчлен
By inspection	Tanlash yo'li bilan	Подбором
Linear factors	Chiziqli ko'paytuvchilar	Линейные множители
Converse theorem	Teskari teorema	Обратная теорема

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

For $x^2 - 7x + 12 = 0$, the sum of the roots is:

For the same equation, the product is:

For $2x^2 + 6x - 8 = 0$, the sum of the roots is:

$x^2 - 5x + 6$ factorises to:

$(x - 2)(x - 3)$

$(x + 2)(x + 3)$

$(x - 1)(x - 6)$

it does not factorise

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the sum of the roots of $x^2 - 4x + 3 = 0$

ANSWER 4

2. Find the product of the roots of $x^2 - 4x + 3 = 0$

ANSWER 3

3. Solve $x^2 - 5x + 6 = 0$ by inspection

ANSWER $x = 2, x = 3$

4. Solve $x^2 - 7x + 12 = 0$ by inspection

ANSWER $x = 3, x = 4$

5. Solve $x^2 + 5x + 6 = 0$ by inspection

ANSWER $x = -2, x = -3$

6. Factorise $x^2 - 3x + 2$

ANSWER $(x - 1)(x - 2)$

7. Factorise $x^2 - 16$

ANSWER $(x - 4)(x + 4)$

Medium · the standard question

7 PROBLEMS

1. Solve $x^2 - 9x + 20 = 0$ by inspection

ANSWER $x = 4, x = 5$

2. Solve $x^2 + x - 12 = 0$ by inspection

ANSWER $x = 3, x = -4$

3. One root of $x^2 + bx - 21 = 0$ is 7. Find the other root and b .

ANSWER $x_2 = -3, b = -4$

4. For $2x^2 + 6x - 8 = 0$, find the sum and product of the roots.

ANSWER -3 and -4

5. Factorise $x^2 - 2x - 15$

ANSWER $(x - 5)(x + 3)$

6. Factorise $3x^2 - 5x - 2$

ANSWER $(x - 2)(3x + 1)$

7. Write a reduced quadratic whose roots are 6 and -2 .

ANSWER $x^2 - 4x - 12 = 0$

Hard · stretch and reason

7 PROBLEMS

1. The roots of $x^2 + px + q = 0$ are 3 and -5 . Find p and q .

ANSWER $p = 2, q = -15$

2. One root of $2x^2 - 9x + c = 0$ is 4. Find c and the other root.

ANSWER Sum = 4.5, so the other root is 0.5 and $c = 2 \cdot 4 \cdot 0.5 = 4$.

3. Factorise $2x^2 + 5x - 12$

ANSWER $(x + 4)(2x - 3)$

4. The roots of $x^2 - 6x + q = 0$ are equal. Find q and the root.

ANSWER $D = 0$ gives $q = 9$ and $x = 3$.

5. For $x^2 - 5x + 6 = 0$, find $x_1^2 + x_2^2$ without solving.

ANSWER $(x_1 + x_2)^2 - 2x_1x_2 = 25 - 12 = 13$

6. For $x^2 + 3x - 10 = 0$, find $\frac{1}{x_1} + \frac{1}{x_2}$ without solving.

ANSWER $\frac{x_1 + x_2}{x_1x_2} = \frac{-3}{-10} = 0.3$

7. Can $x^2 + x + 3$ be factorised into linear factors? Explain.

ANSWER No — $D = 1 - 12 = -11 < 0$, so there are no real roots.

07 Homework

Homework — 6 problems

Algebra 8, Chapter III, pp. 141–146. Use Vieta as a check on every answer.

- Find the sum and product of the roots of $x^2 - 11x + 30 = 0$
- Solve $x^2 - 8x + 15 = 0$ by inspection
- One root of $x^2 + bx - 24 = 0$ is 6. Find the other root and b .
- Factorise $x^2 + 4x - 21$

5. Factorise $2x^2 - 7x + 3$
6. Write a reduced quadratic equation with roots -1 and 8

Biquadratic equations and equations reducible to quadratics

One substitution turns a degree-four equation, or a fractional one, into a quadratic you can already solve.

LESSON 70–71

2 × 40 MIN

ALGEBRA 8, CHAPTER III §24

STAGE 9 · BEYOND

LESSON CLOCK



Warm-up	10 min
Explanation	24 min
Interactive	16 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- Solve a biquadratic equation by the substitution $t = x^2$.
- Reject negative values of t and explain why.
- Solve a fractional-rational equation that reduces to a quadratic.
- Check for values excluded by the denominator.

TEXTBOOK

National	Chapter III, pp. 147–152
Cambridge	Extension beyond Stage 9
Workbook	—

01 Explanation

Biquadratic equations

DEFINITION

An equation of the form $ax^4 + bx^2 + c = 0$ is called **biquadratic**. Only even powers appear.

Put $t = x^2$. Since $x^4 = (x^2)^2 = t^2$, the equation becomes

$$at^2 + bt + c = 0$$

— an ordinary quadratic. Solve it, then **return to x** by solving $x^2 = t$ for each value of t found.

THE STEP EVERYONE FORGETS

$t = x^2$ is a square, so $t \geq 0$. Any **negative** value of t must be **rejected** — it gives no real x . And every positive t gives **two** values of x , namely $\pm\sqrt{t}$.

So a biquadratic equation can have 4, 3, 2, 1 or 0 real roots — count carefully.

A worked pattern

$$x^4 - 13x^2 + 36 = 0$$

$$t^2 - 13t + 36 = 0 \text{ (with } t = x^2\text{)}$$

$$t = 4 \text{ or } t = 9 \text{ (sum 13, product 36)}$$

$$x^2 = 4 \implies x = \pm 2 \quad x^2 = 9 \implies x = \pm 3$$

Four roots: $-3, -2, 2, 3$.

Fractional equations that reduce to quadratics

Multiply through by the common denominator, then solve the quadratic that appears — but list the excluded values **first**.

$$\frac{x}{x-2} + \frac{4}{x} = 3, \quad x \neq 0, x \neq 2$$

Multiplying by $x(x-2)$:

$$x^2 + 4(x-2) = 3x(x-2)$$

$$x^2 + 4x - 8 = 3x^2 - 6x \implies 2x^2 - 10x + 8 = 0 \implies x^2 - 5x + 4 = 0$$

$$x = 1 \text{ or } x = 4 \text{ — both permissible } \checkmark$$

CHECK AGAINST THE EXCLUDED LIST

If a root turns out to be an excluded value it is an **extraneous root** and must be thrown away. Multiplying by an expression that could be zero is exactly what creates them.

02 Worked examples

EXAMPLE 1 Solve $x^4 - 5x^2 + 4 = 0$

① $t = x^2: t^2 - 5t + 4 = 0$

The substitution.

② $t = 1$ or $t = 4$

Sum 5, product 4.

③ $x^2 = 1 \implies x = \pm 1$

④ $x^2 = 4 \implies x = \pm 2$

Four roots in all.

ANSWER

$x = \pm 1, x = \pm 2$

EXAMPLE 2 Solve $x^4 + 3x^2 - 4 = 0$

① $t^2 + 3t - 4 = 0$

② $t = 1$ or $t = -4$

③ $t = -4$ is rejected — a square cannot be negative.

This is the key step.

④ $x^2 = 1 \implies x = \pm 1$

Only two roots.

ANSWER

$x = \pm 1$

EXAMPLE 3 Solve $\frac{6}{x} = x - 1$

① $x \neq 0$

Note the exclusion first.

② $6 = x(x - 1)$

Multiply by x .

③ $x^2 - x - 6 = 0$

④ $x = 3$ or $x = -2$

Neither is 0, so both stand.

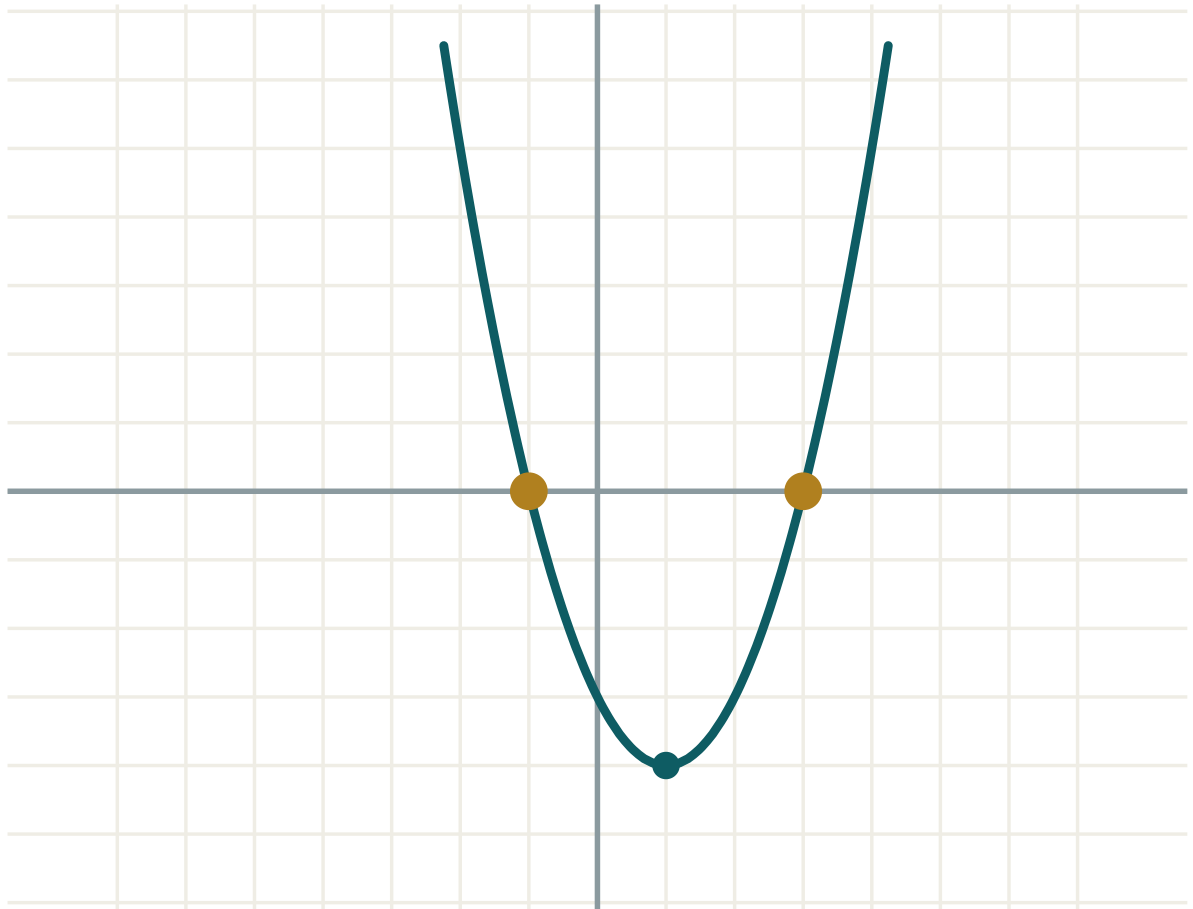
ANSWER

$x = 3, x = -2$

03 Interactive model

Use the model on the quadratic in t , then ask the class how many x -values each t produces.

The quadratic in t

a **1**b **-2**c **-3** $D = b^2 - 4ac$ **16**roots **two** x_1 **-1** x_2 **3** $x_1 + x_2$ **2** $x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Biquadratic equation	Bikvadrat tenglama	Биквадратное уравнение
Substitution	Almashtirish	Замена переменной
New variable	Yangi o'zgaruvchi	Новая переменная
Reject a value	Qiymatni rad etish	Отбросить значение
Fractional-rational equation	Kasr-ratsional tenglama	Дробно-рациональное уравнение
Extraneous root	Chet ildiz	Посторонний корень
Return to x	x ga qaytish	Вернуться к x
Degree of an equation	Tenglamaning darajasi	Степень уравнения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

For a biquadratic the substitution is:

$$t = x$$

$$t = x^2$$

$$t = x^4$$

$$t = \sqrt{x}$$

If $t = -9$ comes out of the substitution, you should:

$$\text{take } x = \pm 3$$

reject it

$$\text{take } x = -3$$

start again

$x^4 - 13x^2 + 36 = 0$ has how many real roots?

2

3

4

0

Before multiplying a fractional equation by its denominator you must:

square both sides

list the excluded values

factorise the numerator

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $x^4 - 5x^2 + 4 = 0$

ANSWER $x = \pm 1, \pm 2$

2. Solve $x^4 - 10x^2 + 9 = 0$

ANSWER $x = \pm 1, \pm 3$

3. Solve $x^4 - 16 = 0$

ANSWER $x = \pm 2$

4. Solve $x^4 - 9x^2 = 0$

ANSWER $x = 0, \pm 3$

5. If $t = x^2$ and $t = 25$, find x .

ANSWER $x = \pm 5$

6. If $t = x^2$ and $t = -1$, find x .

ANSWER No real x .

7. Solve $\frac{4}{x} = 2$

ANSWER $x = 2$

Medium · the standard question

7 PROBLEMS

1. Solve $x^4 + 3x^2 - 4 = 0$

ANSWER $x = \pm 1$

2. Solve $x^4 - 13x^2 + 36 = 0$

ANSWER $x = \pm 2, \pm 3$

3. Solve $x^4 + 5x^2 + 4 = 0$

ANSWER No real roots — both t values are negative.

4. Solve $\frac{6}{x} = x - 1$

ANSWER $x = 3, x = -2$

5. Solve $\frac{12}{x} + x = 7$

ANSWER $x = 3, x = 4$

6. Solve $x^4 - 8x^2 + 16 = 0$

ANSWER $t = 4$ twice, so $x = \pm 2$

7. Solve $\frac{1}{x} + \frac{1}{x+1} = \frac{5}{6}$

ANSWER $x = 2$ or $x = -3$

Hard · stretch and reason

7 PROBLEMS

1. Solve $4x^4 - 17x^2 + 4 = 0$

ANSWER $t = 4$ or $t = 0.25$: $x = \pm 2, \pm 0.5$

2. Solve $\frac{x}{x-2} + \frac{4}{x} = 3$

ANSWER $x = 1, x = 4$

3. Solve $\frac{x+1}{x-1} - \frac{x-1}{x+1} = 1$

ANSWER $4x = x^2 - 1$: $x = 2 \pm \sqrt{5}$

4. Solve $(x^2 - 1)^2 - 5(x^2 - 1) + 4 = 0$

ANSWER Let $t = x^2 - 1$: $t = 1$ or 4 , so $x = \pm\sqrt{2}, \pm\sqrt{5}$

5. For which k does $x^4 + kx^2 + 4 = 0$ have four real roots?

ANSWER Need two positive t -roots: $k^2 > 16$ and $-k > 0$, so $k < -4$.

6. Solve $\frac{2}{x-3} = \frac{x}{x-3} - 1$

ANSWER $x \neq 3$; $2 = x - (x - 3) = 3$ — false, so no solutions.

7. Solve $\frac{x^2 - 4}{x - 2} = 5$

ANSWER $x \neq 2$; $x + 2 = 5$, so $x = 3$.

07 Homework

Homework — 6 problems

Algebra 8, Chapter III, pp. 147–152. Reject negative t values and say so in writing.

1. Solve $x^4 - 26x^2 + 25 = 0$

2. Solve $x^4 + 2x^2 - 3 = 0$

3. Solve $x^4 + 4x^2 + 3 = 0$

4. Solve $\frac{8}{x} = x + 2$

5. Solve $\frac{20}{x} - x = 1$

6. Solve $x^4 - 4x^2 = 0$

Solving problems with quadratic equations

Turning words into a quadratic — and deciding which of the two roots the problem actually allows.

LESSON 72–73

2 × 40 MIN

ALGEBRA 8, CHAPTER III §25

STAGE 9 · 4.1

LESSON CLOCK



Warm-up 10 min

Explanation 24 min

Interactive 16 min

Practice 26 min

Homework 4 min

LEARNING OBJECTIVES

- Choose the unknown and write the equation.
- Solve and interpret both roots in the context.
- Reject a root that has no meaning in the problem.
- Handle the standard types: area, consecutive numbers, speed and work.

TEXTBOOK

National Chapter III, pp. 153–158

Cambridge Learner's Book pp. 84–90

Workbook Workbook 4.1

01 Explanation

The method

1. **Name the unknown.** Write “Let x be ...” and say what unit it is in.
2. **Express everything else** in terms of x .
3. **Write the equation** from the sentence that has not been used yet.
4. **Solve** it.
5. **Check both roots against the problem** and reject any that make no sense.
6. **Answer in words**, with units.

STEP 5 IS THE ONE THAT IS NEW

A quadratic gives two roots, but a length, a speed, a time or a number of people cannot be negative. Say explicitly which root you are rejecting and why — it carries a mark.

The four standard types

Area. A rectangle is 3 m longer than it is wide and has area 40 m^2 . Let the width be x ; then $x(x + 3) = 40$, giving $x = 5$ or $x = -8$. A width cannot be negative, so the

rectangle is 5 m by 8 m .

Consecutive numbers. Two consecutive whole numbers have product 156. Let them be x and $x + 1$: $x^2 + x - 156 = 0$, so $x = 12$ or $x = -13$. Both are valid here — 12 and 13, or -13 and -12 .

Speed. Distance is $s = vt$, so a change of speed on a fixed distance gives a fractional equation that clears to a quadratic.

Pythagoras. A right triangle with one leg x and the other $x + 7$ and hypotenuse 13: $x^2 + (x + 7)^2 = 169$.

A worked speed problem

A cyclist covers 60 km. If the speed were 5 km/h greater, the journey would take one hour less. Find the speed.

Let the speed be x km/h, $x > 0$. The times are $\frac{60}{x}$ and $\frac{60}{x + 5}$, and the first is one hour longer:

$$\frac{60}{x} - \frac{60}{x + 5} = 1$$

$$60(x + 5) - 60x = x(x + 5)$$

$$300 = x^2 + 5x \implies x^2 + 5x - 300 = 0$$

$$x = 15 \text{ or } x = -20 \text{ — a speed cannot be negative}$$

The speed is 15 km/h .

02 Worked examples

EXAMPLE 1 A rectangle is 3 m longer than it is wide and has area 40 m². Find its dimensions.

① Let the width be x m; the length is $x + 3$ m.

② $x(x + 3) = 40$
Area.

③ $x^2 + 3x - 40 = 0$

④ $x = 5$ or $x = -8$

⑤ A width cannot be negative, so $x = 5$.
Reject the other root.

ANSWER

5 m by 8 m

EXAMPLE 2 The product of two consecutive whole numbers is 156. Find them.

① Let them be x and $x + 1$.

② $x(x + 1) = 156$

③ $x^2 + x - 156 = 0$

④ $x = 12$ or $x = -13$
Both give whole numbers here.

ANSWER

12 and 13, or -13 and -12

EXAMPLE 3 A cyclist covers 60 km. Riding 5 km/h faster would save an hour. Find the speed.

① Let the speed be x km/h.

②
$$\frac{60}{x} - \frac{60}{x+5} = 1$$

The time saved.

③
$$x^2 + 5x - 300 = 0$$

Multiply by $x(x+5)$.

④
$$x = 15 \text{ or } x = -20$$

Reject the negative speed.

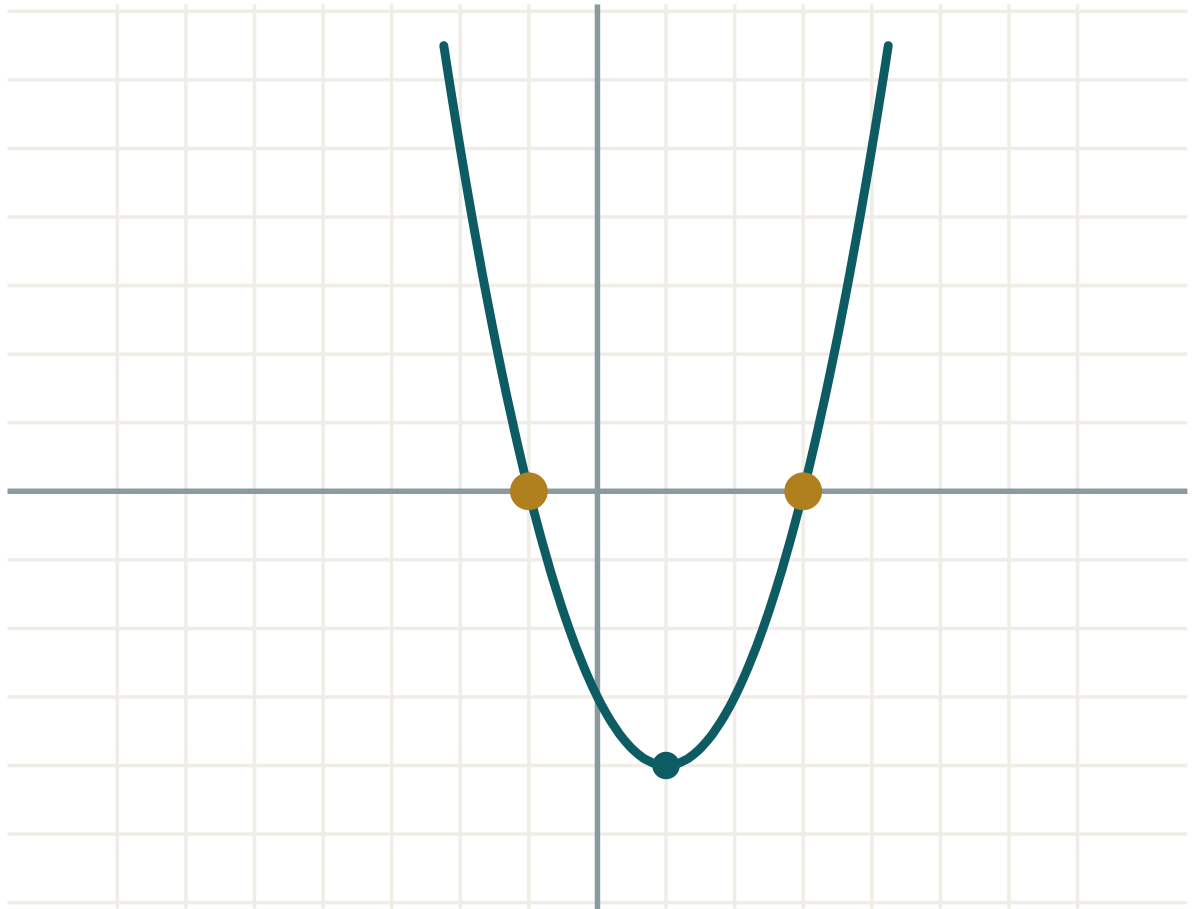
ANSWER

15 km/h

03 Interactive model

Write the equation on the board and let the model show which root the graph gives.

The equation behind the problem

a **1**b **-2**c **-3** $D = b^2 - 4ac$ **16**roots **two** x_1 **-1** x_2 **3** $x_1 + x_2$ **2** $x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Word problem	Matnli masala	Текстовая задача
Let x be...	x deb belgilaymiz	Пусть x —
Set up an equation	Tenglama tuzish	Составить уравнение
Reject a root	Ildizni rad etish	Отбросить корень
Consecutive numbers	Ketma-ket sonlar	Последовательные числа
Average speed	O'rtacha tezlik	Средняя скорость
Dimensions	O'lchamlar	Размеры
Interpret the answer	Javobni izohlash	Истолковать ответ

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A rectangle has width x and length $x + 4$ and area 45. The equation is:

A length comes out as $x = -6$ or $x = 4$. You should:

Two consecutive numbers are written as:

Time equals:

$$\text{distance} \times \text{speed}$$

$$\frac{\text{distance}}{\text{speed}}$$

$$\frac{\text{speed}}{\text{distance}}$$

$$\text{distance} + \text{speed}$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *A square has area 81 cm^2 . Find its side.*

ANSWER 9 cm

2. *A number squared is 49. Find it, given it is positive.*

ANSWER 7

3. *Two consecutive numbers have product 20. Find them.*

ANSWER 4 and 5 (or -5 and -4)

4. *A rectangle is 2 m longer than wide, area 24 m^2 . Write the equation.*

ANSWER $x(x + 2) = 24$

5. *Solve that equation.*

ANSWER $x = 4$, so $4 \text{ m} \times 6 \text{ m}$

6. *A number added to its square gives 12. Find the positive number.*

ANSWER 3

7. *The area of a square is 144. Find the perimeter.*

ANSWER 48

Medium · the standard question

7 PROBLEMS

1. A rectangle is 3 m longer than wide and has area 40 m^2 . Find its dimensions.

ANSWER $5 \text{ m} \times 8 \text{ m}$

2. The product of two consecutive whole numbers is 156. Find them.

ANSWER 12 and 13

3. A right triangle has legs x and $x + 7$ and hypotenuse 13. Find the legs.

ANSWER 5 and 12

4. A number is 3 less than its square. Find both values.

ANSWER $x^2 - x - 3 = 0$: $x = \frac{1 \pm \sqrt{13}}{2}$

5. The sum of a number and its reciprocal is 2.5. Find the number.

ANSWER $x = 2$ or $x = 0.5$

6. A rectangle has perimeter 26 cm and area 40 cm^2 . Find its sides.

ANSWER 5 cm and 8 cm

7. Two consecutive even numbers have product 168. Find them.

ANSWER 12 and 14

Hard · stretch and reason

7 PROBLEMS

1. A cyclist covers 60 km. Riding 5 km/h faster would save an hour. Find the speed.

ANSWER 15 km/h

2. A train covers 240 km. Going 20 km/h faster would save one hour. Find the speed.

ANSWER $x^2 + 20x - 4800 = 0$: 60 km/h

3. A rectangular garden 20 m by 15 m has a path of uniform width around it, doubling the area. Find the width.

ANSWER $(20 + 2w)(15 + 2w) = 600$: $w = 2.5$ m

4. The sum of the squares of two consecutive whole numbers is 313. Find them.

ANSWER $x^2 + (x + 1)^2 = 313$: 12 and 13

5. A stone is thrown up and its height is $h = 20t - 5t^2$. When is it 15 m high?

ANSWER $5t^2 - 20t + 15 = 0$: at $t = 1$ s going up and $t = 3$ s coming down.

6. Two workers together finish a job in 6 days. Alone, one takes 5 days longer than the other. Find both times.

ANSWER $\frac{1}{x} + \frac{1}{x+5} = \frac{1}{6}$: 10 and 15 days.

7. The area of a triangle is 30 cm² and its height is 7 cm more than its base. Find the base.

ANSWER $\frac{1}{2}x(x + 7) = 30$: $x = 5$ cm

07 Homework

Homework — 6 problems

Algebra 8, Chapter III, pp. 153–158. Always say which root you reject and why.

1. A rectangle is 5 cm longer than wide and has area 84 cm². Find its dimensions.
2. Two consecutive whole numbers have product 342. Find them.
3. A right triangle has legs x and $x + 2$ and hypotenuse 10. Find the legs.
4. A car covers 180 km. Going 10 km/h faster would save half an hour. Find the speed.

5. *The sum of a number and its square is 56. Find both values.*
6. *A rectangle has perimeter 34 m and area 60 m². Find its sides.*

Compound percentages, simultaneous equations and quadratic graphs

The national “practical and interdisciplinary” lessons, carrying three Cambridge Stage 9 topics the programme does not otherwise reach.

LESSON 74–76

3 × 40 MIN

ALGEBRA 8, PRACTICAL PROBLEMS

STAGE 9 · 3.3, 4.2, 10.2 [CAMBRIDGE INSERT]

LESSON CLOCK

15'

36'

24'

39'

6'

Warm-up	15 min
Explanation	36 min
Interactive	24 min
Practice	39 min
Homework	6 min

LEARNING OBJECTIVES

- Calculate repeated percentage change with a multiplier.
- Solve a pair of simultaneous linear equations by substitution and by elimination.
- Plot the graph of a quadratic and read its roots and vertex.
- Apply all three to practical situations.

TEXTBOOK

National	pp. 159–164
Cambridge	Learner’s Book pp. 66–73, 91–95, 218–224
Workbook	Workbook 3.3, 4.2, 10.2

01

Explanation

Lesson 74 — compound percentage change

THE MULTIPLIER

An increase of $p\%$ multiplies by $1 + \frac{p}{100}$; a decrease multiplies by $1 - \frac{p}{100}$. For n repetitions, raise the multiplier to the power n :

$$final = initial \times (multiplier)^n$$

A salary of 2 000 000 so’m rising 8% a year for 3 years: $2\,000\,000 \times 1.08^3 \approx 2\,519\,424$.

NOT THE SAME AS ADDING THE PERCENTAGES

Three years at 8% is **not** 24%. It is $1.08^3 \approx 1.2597$, that is 25.97% — the extra comes from the growth compounding on itself.

A 20% rise followed by a 20% fall is $1.2 \times 0.8 = 0.96$ — a 4% **loss**, not a return to the start. This is the single most useful fact in the topic.

Lesson 75 — simultaneous equations

Substitution. Make one unknown the subject of one equation and put it into the other.

$$y = 2x + 1 \text{ and } 3x + y = 11 \implies 3x + 2x + 1 = 11 \implies x = 2, y = 5$$

Elimination. Add or subtract the equations so that one unknown disappears — multiplying one or both first if you need matching coefficients.

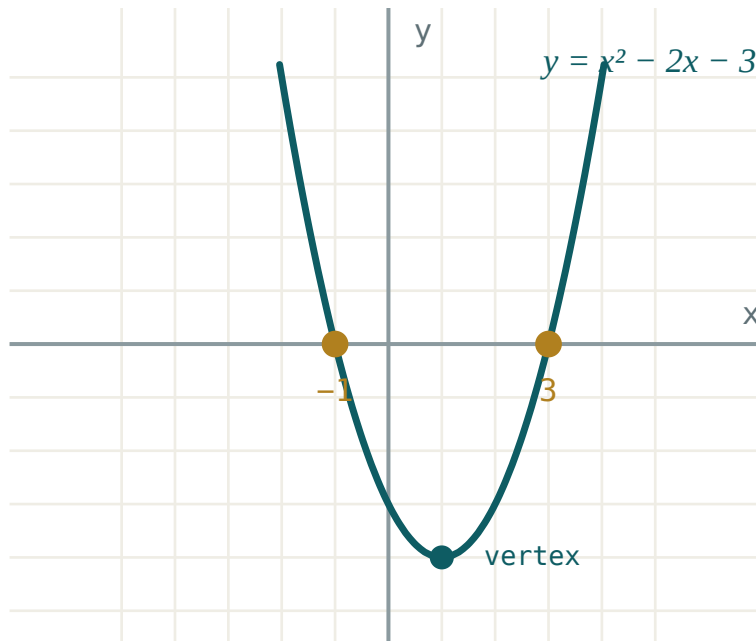
$$2x + 3y = 12 \text{ and } 2x - y = 4: \text{ subtract } \implies 4y = 8 \implies y = 2, x = 3$$

Graphically the solution is where the two lines cross. Parallel lines mean no solution; the same line twice means infinitely many.

Always **check in both equations** — a pair that satisfies only one is not a solution.

Lesson 76 — the graph of a quadratic

To draw $y = ax^2 + bx + c$, build a table of values, plot, and join with a smooth curve — never with straight segments.



The parabola crosses the x -axis at the roots and turns at the vertex.

WHAT TO READ OFF THE GRAPH

- **Roots** — where the curve meets the x -axis.
- **y -intercept** — the value c .
- **Axis of symmetry** — the vertical line $x = -\frac{b}{2a}$.
- **Vertex** — on that line; the lowest point if $a > 0$, the highest if $a < 0$.

$a > 0$ gives a curve opening upwards (a “valley”); $a < 0$ opens downwards (a “hill”).

The graph is a picture of everything the discriminant told you.

02 Worked examples

EXAMPLE 1 A population of 50 000 grows 3% a year. Find it after 4 years.

① $multiplier = 1.03$

A 3% increase.

② $50\,000 \times 1.03^4$

Four years.

③ $= 50\,000 \times 1.1255$

④ $\approx 56\,275$

ANSWER

$\approx 56\,300$ people

EXAMPLE 2 Solve $3x + 2y = 16$ and $x - y = 3$

① $x = y + 3$

Make x the subject of the second.

② $3(y + 3) + 2y = 16$

Substitute.

③ $5y = 7, y = 1.4$

④ $x = 4.4$

Check: $3(4.4) + 2(1.4) = 16$ ✓

ANSWER

$x = 4.4, y = 1.4$

EXAMPLE 3 For $y = x^2 - 2x - 3$, find the roots, the y -intercept and the vertex.

① $x^2 - 2x - 3 = 0 \implies (x - 3)(x + 1) = 0$

Roots 3 and -1 .

② y -intercept: $c = -3$

Put $x = 0$.

③ axis: $x = -\frac{-2}{2} = 1$

④ $y = 1 - 2 - 3 = -4$

Vertex $(1; -4)$.

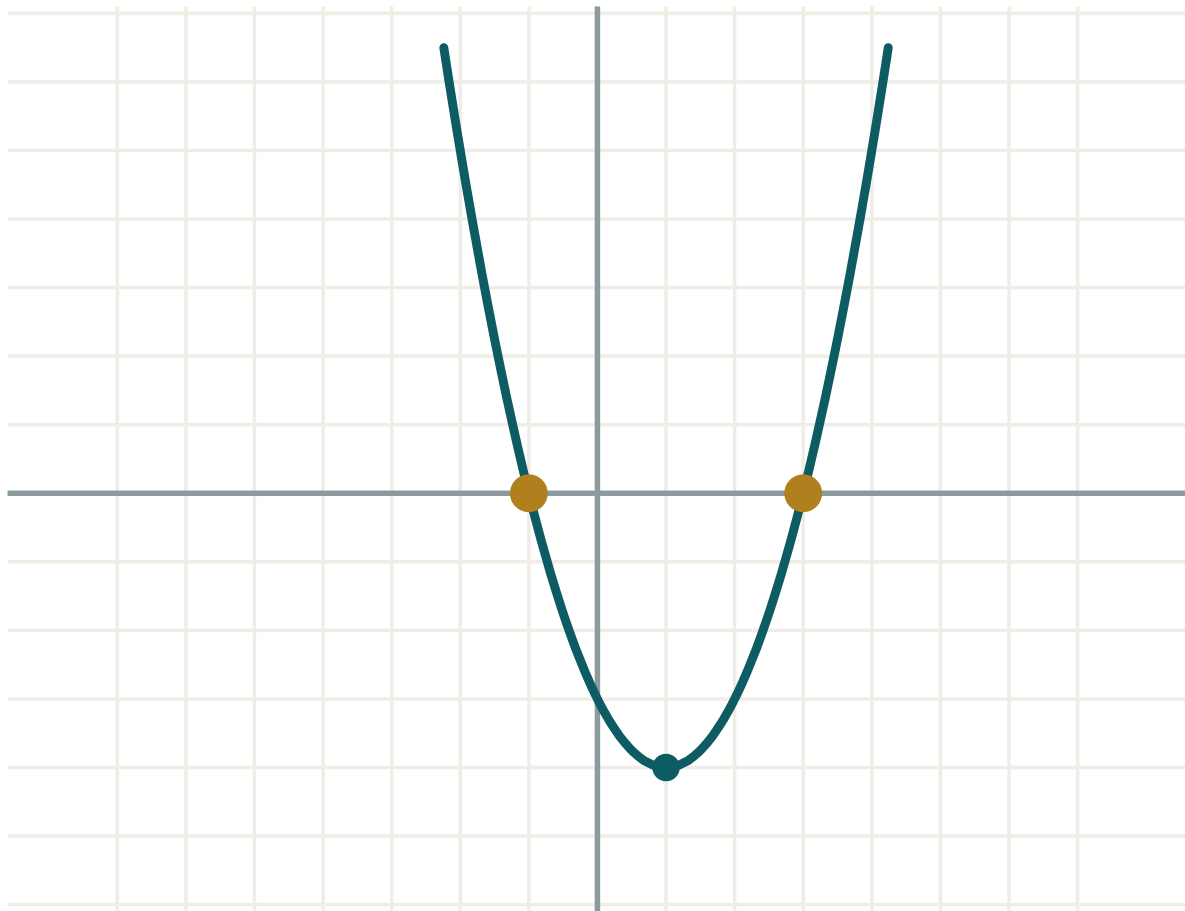
ANSWER

Roots $-1, 3$; intercept -3 ; vertex $(1; -4)$

03 Interactive model

Change a , b and c and let the class predict the vertex before the readout shows it.

Reading a parabola



a 1

b -2

c -3

 $D = b^2 - 4ac$ 16

roots two

 x_1 -1 x_2 3 $x_1 + x_2$ 2 $x_1 \cdot x_2$ -3

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Compound percentage	Murakkab foiz	Сложный процент
Multiplier	Ko'paytiruvchi	Множитель
Increase / decrease	Ortish / kamayish	Увеличение / уменьшение
Simultaneous equations	Tenglamalar sistemasi	Система уравнений
Substitution method	O'rniga qo'yish usuli	Метод подстановки
Elimination method	Qo'shish usuli	Метод сложения
Table of values	Qiymatlar jadvali	Таблица значений
Vertex of a parabola	Parabola uchi	Вершина параболы
Axis of symmetry	Simmetriya o'qi	Ось симметрии

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A 10% rise then a 10% fall leaves you with:

the same

1% more

1% less

10% less

Three years at 5% multiplies by:

1.15

1.05^3

3×1.05

0.95^3

The axis of symmetry of $y = x^2 - 6x + 5$ is:

$x = 3$

$x = -3$

$x = 6$

$x = 5$

Two simultaneous equations whose lines are parallel have:

one solution

no solution

two solutions

infinitely many

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Increase 200 by 10%*

ANSWER 220

2. *Decrease 200 by 10%*

ANSWER 180

3. *Write the multiplier for a 7% increase*

ANSWER 1.07

4. *Write the multiplier for a 15% decrease*

ANSWER 0.85

5. *Solve $x + y = 7$ and $x - y = 1$*

ANSWER $x = 4, y = 3$

6. *Solve $y = 2x$ and $x + y = 9$*

ANSWER $x = 3, y = 6$

7. *Find the y-intercept of $y = x^2 + 3x - 4$*

ANSWER -4

Medium · the standard question

7 PROBLEMS

1. A population of 50 000 grows 3% a year. Find it after 4 years.

ANSWER $\approx 56\,300$

2. A car worth 30 000 000 so'm loses 12% a year. Find its value after 3 years.

ANSWER $\approx 20\,440\,000$

3. A 20% rise is followed by a 20% fall. Find the overall change.

ANSWER 4% decrease

4. Solve $3x + 2y = 16$ and $x - y = 3$

ANSWER $x = 4.4, y = 1.4$

5. Solve $2x + 3y = 12$ and $2x - y = 4$

ANSWER $x = 3, y = 2$

6. Find the roots and vertex of $y = x^2 - 2x - 3$

ANSWER Roots $-1, 3$; vertex $(1; -4)$

7. Find the axis of symmetry of $y = 2x^2 - 8x + 1$

ANSWER $x = 2$

Hard · stretch and reason

7 PROBLEMS

1. A sum grows to 1.331 times its value in 3 years at a constant rate. Find the rate.

ANSWER $r^3 = 1.331$, so $r = 1.1$ — 10% a year.

2. A price falls 25% then rises 25%. Find the overall change.

ANSWER $0.75 \times 1.25 = 0.9375$ — a 6.25% loss.

3. Solve $4x - 3y = 11$ and $3x + 2y = 4$

ANSWER $x = 2, y = -1$

4. Two numbers add to 30 and differ by 8. Find them, using simultaneous equations.

ANSWER 19 and 11

5. Sketch $y = -x^2 + 4x$ and give its roots and vertex.

ANSWER Roots 0 and 4; vertex (2; 4); opens downwards.

6. For which values of c does $y = x^2 - 4x + c$ touch the x -axis?

ANSWER $D = 0$ gives $c = 4$.

7. A shop raises a price by 10% each year for 2 years, then cuts it by 21%. Show the price returns to the start.

ANSWER $1.1^2 \times 0.79 = 1.21 \times 0.79 \approx 0.956$ — close, but a 4.4% loss; an exact return needs a cut of 17.36%.

07 Homework

Homework — 6 problems

Algebra 8, pp. 159–164, and Cambridge Learner's Book 3.3, 4.2, 10.2.

1. A town of 80 000 grows 2% a year. Find its population after 5 years.
2. A machine worth 12 000 000 so'm loses 15% a year. Find its value after 4 years.
3. Solve $5x + 2y = 21$ and $3x - y = 4$
4. Solve $2x + y = 10$ and $x - 3y = -9$

5. Draw the graph of $y = x^2 - 4x + 3$ for x from -1 to 5 , and read off the roots.
6. Find the vertex and the axis of symmetry of $y = x^2 + 6x + 5$

Control work 6 · Quadratic equations

The Chapter III assessment, and the four errors it produces every year.

LESSON 77–78

2 × 40 MIN

ALGEBRA 8, CHAPTER III

STAGE 9 · 4.1

LESSON CLOCK

4'	72'	4'
Setting up	4 min	
The paper	72 min	
Collect in	4 min	

LEARNING OBJECTIVES

- Assess incomplete equations, the discriminant, the formula, Vieta and word problems.
- Diagnose each lost mark by error type.
- Re-solve every task that was lost.

TEXTBOOK

National	Revision of Chapter III
Cambridge	Learner's Book pp. 84–95
Workbook	Workbook 4.1

01 Explanation

Lesson 77 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each:

- task 1 · an incomplete quadratic
- task 2 · solve by factorising
- task 3 · find D and say how many roots
- task 4 · solve with the formula
- task 5 · Vieta — find a missing coefficient or root
- task 6 · a biquadratic equation
- task 7 · a word problem

Marking note: task 7 loses a mark if the impossible root is not rejected in writing.

Lesson 78 — work on mistakes (40 minutes)

THE FOUR ERRORS THIS PAPER PRODUCES

1. **Sign slips in D** — writing $-4ac$ as negative when c is already negative.
2. **Dividing by x** , losing the root $x = 0$.
3. **Negative t kept** in a biquadratic, producing imaginary “roots”.

4. The impossible root not rejected in a word problem.

02 Worked examples

EXAMPLE 1 Find the error: for $x^2 - 2x - 3 = 0$, $D = 4 - 12 = -8$

① $c = -3$, so $-4ac = -4 \cdot 1 \cdot (-3) = +12$.

Two minus signs make a plus.

② $D = 4 + 12 = 16$

③ $x = \frac{2 \pm 4}{2} = 3 \text{ or } -1$

ANSWER

$D = 16$, roots 3 and -1

EXAMPLE 2 Find the error: $x^2 = 4x$, so $x = 4$

① Both sides were divided by x — which may be zero.

② $x^2 - 4x = 0 \implies x(x - 4) = 0$

Factorise instead.

③ $x = 0$ or $x = 4$

The root 0 was lost.

ANSWER

$x = 0$, $x = 4$

03 Interactive model

Show the wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: for $x^2 - 2x - 3 = 0$, $D = 4 - 12 = -8$

① $a = 1, b = -2, c = -3$

Read
the
signs
off
carefully.

② $-4ac = -4 \cdot 1 \cdot (-3) = +12$

The
two
minus
signs
cancel.

③ $D = 4 + 12 = 16 > 0$

Two
roots,
not
none.

④ $x = 3$

or $x = -1$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Discriminant	Diskriminant	Дискриминант
Root	Ildiz	Корень
Reject	Rad etish	Отбросить
Check by substitution	O'rniga qo'yib tekshirish	Проверка подстановкой
Word problem	Matnli masala	Текстовая задача

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

For $x^2 - 2x - 3 = 0$, D is:

-8

16

4

-4

$x^2 = 4x$ has roots:

$x = 4$

$x = 0, 4$

$x = \pm 2$

$x = 0$

In a biquadratic, $t = -4$ gives:

$x = \pm 2$

no real x

$x = 2$

$x = -2$

A width comes out as 5 or -8 . The answer is:

both

-8

5, rejecting -8

neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** Solve $x^2 - 36 = 0$

ANSWER $x = \pm 6$

2. **Task 2.** Solve $x^2 - 6x + 8 = 0$

ANSWER $x = 2, x = 4$

3. **Task 3.** Find D for $x^2 + 2x + 5 = 0$ and say how many roots.

ANSWER $D = -16$, none

4. **Task 4.** Solve $2x^2 - 5x + 2 = 0$

ANSWER $x = 2, x = 0.5$

5. **Task 5.** One root of $x^2 + bx - 12 = 0$ is 4. Find b .

ANSWER other root -3 , $b = -1$

6. **Task 6.** Solve $x^4 - 5x^2 + 4 = 0$

ANSWER $x = \pm 1, \pm 2$

7. **Task 7.** A rectangle is 2 m longer than wide, area 35 m². Find its sides.

ANSWER $5\text{ m} \times 7\text{ m}$

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** Solve $3x^2 - 12x = 0$

ANSWER $x = 0, x = 4$

2. **Task 2.** Solve $x^2 + x - 20 = 0$

ANSWER $x = 4, x = -5$

3. **Task 3.** Find D for $2x^2 - 4x + 2 = 0$ and say how many roots.

ANSWER $D = 0$, one

4. **Task 4.** Solve $3x^2 + 5x - 2 = 0$

ANSWER $x = -\frac{1}{3}, x = -2$

5. **Task 5.** The roots of $x^2 + px + q = 0$ are 2 and -7 . Find p and q .

ANSWER $p = 5, q = -14$

6. **Task 6.** Solve $x^4 - 10x^2 + 9 = 0$

ANSWER $x = \pm 1, \pm 3$

7. **Task 7.** Two consecutive whole numbers have product 210. Find them.

ANSWER 14 and 15

Hard · stretch and reason

7 PROBLEMS

1. Find the error: for $x^2 - 2x - 3 = 0$, $D = 4 - 12 = -8$

ANSWER $-4ac = +12$; correct $D = 16$, roots 3, -1 .

2. Find the error: $x^2 = 4x$, so $x = 4$

ANSWER Dividing by x lost the root 0.

3. Find the error: $x^4 + 3x^2 - 4 = 0$ gives $x = \pm 1, \pm 2$

ANSWER $t = -4$ must be rejected; only $x = \pm 1$.

4. Find the error: a width comes out 5 or -8 , answer “5 and -8 ”

ANSWER A length cannot be negative — the rejection must be stated.

5. Find the error: “ $x^2 = 16$, so $x = 4$ ”

ANSWER The negative root was lost: $x = \pm 4$.

6. Find the error: “ $(x - 1)(x - 4) = 6$, so $x - 1 = 6$ or $x - 4 = 6$ ”

ANSWER The zero product rule needs a zero on the right; expand and re-solve.

7. Find the error: “ $D = 25$, so $x = \frac{-b \pm 25}{2a}$ ”

ANSWER The formula uses $\sqrt{D} = 5$, not D itself.

07 Homework

After the work-on-mistakes lesson

Re-solve every task you lost marks on. Quarter IV opens with statistics.

- Solve $2x^2 - 7x - 4 = 0$
- Solve $x^4 - 13x^2 + 36 = 0$
- One root of $x^2 + bx + 15 = 0$ is 5. Find b and the other root.
- A rectangle is 4 cm longer than wide and has area 96 cm^2 . Find its sides.

5. Write out the four errors from this lesson with one example each.